



The prosody of the extended VP

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Abstract

This paper investigates the syntax of VSO and VOS clauses in Mandar (Austronesian) by leveraging the prosody. This language allows free alternations between VSO and VOS orders, but phonotactic diagnostics reveal that VSO strings are optimally parsed into tight prosodic constituents while VOS strings are not. These results converge with syntactic diagnostics to show that VSO orders arise from leftward movement of the verb while VOS orders are generated through an additional step of rightward $\bar{A}$ -scrambling of the S. Targeted manipulations then reveal that phonological phrases can be built around substrings of arguments in the VSO string, providing a new type of evidence for fine functional structure in the extended VP (Larson 1988). The prosodic parse of scrambled arguments, finally, shows that $\bar{A}$ -scrambling places its targets in adjunct positions (Chomsky 1993) in Mandar, setting up an account of scrambling that is grounded in the principle of Greed (Lasnik 1995).

Keywords Prosody · VP-structure · Verb-initiality · Scrambling · Sulawesi

1 Introduction

The internal structure of the verb phrase has risen to become a central domain of syntactic inquiry since the initial discovery of functional structure therein (Larson 1988). Work in this space typically begins from a question of constituency: how are the verb and its arguments arranged hierarchically in the extended VP? Work on this question has come to guide many fundamental lines of research in syntactic theory, on the systems that yield linear order (Sproat 1985; McCloskey 1991; Massam 2001a; Otsuka 2005), the constraints that govern binding and coreference (Ernst 1994; Bruening 2014), the derivational course of structure building (Phillips 2003), and the theories of selection, adjunction, and reprojection (Landau 2007a; Janke and Neeleman 2012). Given the importance of this work, then, it is a valuable task to extend and refine the analytical toolkit that lets us study the structure of the VP.

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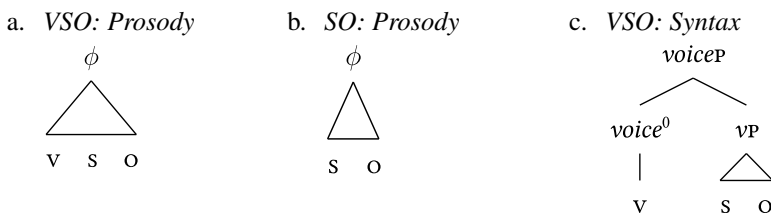
This paper lays the foundations to study the syntax of the VP in a new way. On an empirical level, its focus falls on the syntax of VSO-VOS alternations in Mandarin, an Austronesian language of Indonesia. VSO clauses are often assumed to show a right-branching or *descending* syntax, on which the V C-commands the following S and O (1a). But VOS strings often raise puzzles of constituency: do these strings show the same type of descending syntax, with the V C-commanding the following S (1b), or do they show a left-branching or *ascending* syntax, where the V is C-commanded by the S to its right (1c)?

(1) *The Syntax of VSO and VOS*

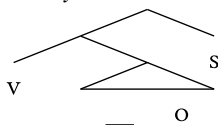
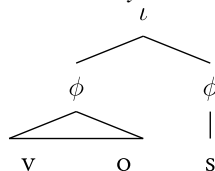
- a. VSO: DESCENDING SYNTAX: [_{voiceP} V [_{VP} S [_{VP} _ O]]]
- b. VOS: DESCENDING SYNTAX: [_{voiceP} V [_{FP} O [_{VP} S [_{VP} _]]]]
- c. VOS: ASCENDING SYNTAX: [_{FP} [_{voiceP} V [_{VP} _ [_{VP} _ O]]]] S]

To study the syntax of VSO and VOS strings in Mandarin, this paper will turn to the prosody. The theory of the Prosodic Hierarchy holds that phonological strings are organized into hierarchical constituent structures that correspond in a systematic but imperfect way to the syntax beneath them (Selkirk 1984, 1986; Nespor and Vogel 1986). The abstract phonological constituents that make up this structure can be detected from the distribution of phonotactic restrictions in the phrasal phonology, and this fact will allow us to build a properly phonological case for the syntactic constituency of the postverbal space. Focusing on the optimal prosody that emerges when well-planned clauses are produced under broad focus, we can see two things in Mandarin clauses of the order VSO. First, the string VSO typically forms a phrase-level prosodic constituent (2a). Second, under the right circumstances, the substring SO does as well (2b). These results lay the foundation for a particular analysis of the syntax: one on which the VSO string forms a constituent that is built by raising of the V out of a smaller projection that also contains the S and O (2c).

(2) *Prosodic Evidence for Functional Structure*



The same investigation of syntax and prosody will ultimately lead to a separate conclusion about Mandarin strings of the order VOS: the derivation of these strings must also involve a process of rightward $\bar{A}$ -scrambling that shifts the S to a position above the VO string (3a). The finer prosody of these clauses, shown in (3b), then leads to a final syntactic result: the relevant type of scrambling must place its targets in adjunct positions. This observation, in turn, opens up a properly phonological case that, at least in Mandarin, $\bar{A}$ -scrambling must result in adjunction (Chomsky 1993)—and thus cannot be driven by the logic of Attract (Lasnik 1995; Bošković 1995).

(3) *Prosodic Evidence for Rightward Scrambling*a. *The Syntax: VOS*b. *The Prosody: VOS*

The remainder of the paper is structured as follows. Section 2 provides background information on Mandar and the methodology of elicitation, lays out the facts of word order, introduces the theory of the prosodic hierarchy, and sets up diagnostics for the maximal phonological phrase. Section 3 presents the prosodic organization of VSO strings and supplements those findings with syntax-internal investigation to develop a theory of Mandar clause structure. Section 4 lays out a strategy to create specific mismatches between syntactic and prosodic constituency and then leverages it to draw out phonological evidence for two functional projections in the *voice*P: *v*P and *appl*P. Section 5 turns to the prosodic organization of VOS clauses and works through the prosody of adjunction to build the scrambling analysis. Section 6 concludes by drawing connections to the emerging phonology of selection.

2 Background

Mandar is an Austronesian language spoken by 400,000 people in West Sulawesi, Indonesia (Grimes and Grimes 1987). It is a member of the South Sulawesi subfamily and has been the subject of an Indonesian-language grammar (Pelenkahu et al. 1983), a description of adverbs (Sikki et al. 1987), a compilation of traditional poetry (Muthalib and Sangi 1991), a conversational handbook (Friberg and Jerniati 2000), and many smaller works by the Language Office of South Sulawesi (e.g., Jerniati 2005, 2017). This paper will focus on the standard variety of the language, spoken along the coastal plain between the cities of Polewali and Majene.¹²

Mandar is a predicate-initial language that shows flexible word order in the postverbal domain. The language allows for *pro*-drop and shows no morphological case. Transitive verbs host ergative agreement with the external argument and every finite clause contains a second-position clitic that spells out absolutive agreement. These features are shown in the Mandar clause in example (4) below.

¹ GLOSSING: ABS: absolutive, AF: agent focus ANTIP: antipassive, APPL: applicative, ERG: ergative, COM: comitative; GEN: genitive; 1SG: first-person singular. ORTHOGRAPHY: <c>: /tʃ/, <'>: /ʔ/, <ng>: /ŋ/.

² Mandar can also be compared with the other languages of the South Sulawesi subfamily, whose syntax is described by, among others, Friberg (1991, 1996), Matti (1994), Strømme (1994), Valkama (1995a,b), Jukes (2006), Kaufman (2008), Lee (2008), Laskowske (2016), and especially Finer (1994, 1997, 1998, 1999), Béjar (1999); and Finer and Basri (2020).

(4) *The Shape of a Mandar Clause*

Na-solangang a' yau iKaco' mua' lamba a' *pro* manini.
 3ERG-accompany 1ABS 1SG NAME if go 1ABS later

'Kacho' will accompany me if I go later.'

This investigation of Mandar will focus on the phrasal phonology of the language, and more specifically, on phonotactic restrictions that emerge at the edges of particular prosodic domains. These processes interact in crisp and categorical ways with each other and with the system of stress, they are easy to hear in regular speech, and they are often reflected in the popular orthography. Nevertheless, they are difficult to study for two reasons. First, they are sensitive to the phonology of focus and to many non-phonological factors that can derail the default patterns of prosodic phrasing, like hesitations in planning. Second, they are often deemed substandard in untrained elicitation contexts by speakers who wish to emphasize the ways that words are pronounced in isolation. In my experience, then, these processes can only be studied in working relationships that are relatively deep, in which consultants have internalized the goals of linguistic investigation, accepted the existence of phrasal phonology as a real component of the grammar, become accustomed to producing naturalistic broad-focus prosody in elicitation contexts, and ultimately learned to give judgments about the optimal application of rules in the phrasal phonology.

The judgments in this study have thus been gathered over two years of targeted work on prosodic phonology with one speaker of Mandar, Jupri Talib, with whom I have been working for five years overall. The processes under investigation were first noted through natural observation and their distribution was documented slowly, as the influence of focus was sorted out, as Jupri was taught the essentials of Prosodic Hierarchy Theory, and as a strategy was developed to study prosody through pairwise comparison. To collect each judgment, Jupri was given a pair of written sentences that differed in the application of a phonological rule. Once the sentences were established as pre-phonologically acceptable, he was then walked through a context in which the sentences could be uttered under broad focus. After that, he was asked to produce each sentence several times. Once the sentences had become familiar, Jupri was then asked whether one version could be identified as acceptable and comparatively "more normal" (Indonesian: *lebih biasa*) in contexts that did not place narrow focus on any constituent (*konteks biasa*). If this was possible, he was then asked to rate the second example as "also possible with regular prosody" (*bisa juga dengan irama biasa*), "only possible with narrow focus" (*hanya dengan penekanan khusus*), or "incorrect" (*salah*). In the summer of 2022, this strategy was repeated with several item sets to test every generalization in this paper; in the summer of 2023 it was repeated at a similar level of scrutiny with different examples and yielded identical results. The primary language of contact was Indonesian; Mandar was used often as well.³

At the end of the research process, several recordings were made with Jupri; these are presented in the appendix. A subset of the data was then rechecked with Hairuddin

³This discussion will not engage with the phonology of focus, which consistently derails the regular mapping from syntax to prosody in Mandar. The core generalizations about contrastive focus in Mandar are: (i) Contrastive foci do not occupy a unique linear position (though two kinds of focus-movement exist); (ii) Maximal phonological phrase boundaries are inserted after contrastive foci, wherever they fall, and

(Universitas Hasanuddin) and Sitti Sapia (Universitas Sulawesi Barat), two native-speaker phonologists; both offered judgments that were consistent with Jupri's. Based on these results, I believe that the following description of Jupri's idiolect is a replicable representation of the standard variety of Mandar.

2.1 The ordering alternation

The following diagram shows the linear position of the Mandar verb: across all clause types, it follows complementizers, negation, and auxiliaries, precedes VP-adjuncts, and precedes arguments that are not moved left by topicalization or WH-movement.

(5) *The Shape of the Mandar Clause*

C > NEG > AUX > V > ARGUMENTS & ADJUNCTS

The following example shows this order: the V *pole* 'come' appears after negation and an auxiliary and precedes the adjunct *dini* 'here' and the subject *Kacho* (a name with the proprial article *i*). The same order emerges in the following non-finite clause, where the predicate *karambo* 'far' precedes the subject *boyanna* 'his house.'⁴

(6) *The Position of the Verb*

Ndammi rua pole dini [_S iKaco'] tappa-na karambo boyan-na.
not.3ABS have come here NAME since-3GEN far house-3GEN

'Kacho' hasn't come here since his house became far away.'

Sikki et al. (1987, 316)

I argue that there is a consistent basic order of postverbal arguments in Mandar: EXTERNAL ARGUMENT > INTERNAL ARGUMENT > APPLIED ARGUMENT (7). Using terminology that avoids reference to Case, I will refer to the external argument as the S, the internal argument as the O, and the applied argument as the D. I will also represent VP-final adjuncts as X. In these terms, example (7) shows the order VSODX.

(7) *The Order VSODX*

Na-alli-ang i [_S iKaco'] [_O bunga] [_D iCicci'] [_X dio].
3ERG-buy-APPL 3ABS NAME flower NAME there

'Kacho' bought a flower for Chichi' there.'

In all the clause types studied below, the order VSOD will always be possible. Above, example (7) shows that this order is possible in ditransitive clauses

(iii) In the space before contrastive foci, all expected maximal phonological phrase boundaries are erased (so that contrastive foci are final in the first maximal phonological phrase of the *i*). In the space after contrastive foci, all expected maximal phonological phrase boundaries appear undisturbed.

⁴This paper will set aside the syntax of non-verbal predication, which differs from verbal predication in Mandar in many ways. Notably, verbal and non-verbal predicates are distinguished in Mandar in the prosody: verbal predicates are typically parsed into larger ϕ s with following arguments when phonological constraints allow, but non-verbal predicates are not parsed into ϕ s with following arguments in this way.

when the O is indefinite; example (8a) below shows that it is also possible in ditransitive clauses where the O is definite. The related order VSO, in turn, will always be possible in clauses that lack a D. Example (8b) illustrates this second fact: the order VSO is possible in clauses that contain transitive verbs, where the O is definite and absolutive (the absolutive argument must always be definite in Mandarin).

(8) *VSOD Order: Insensitive to Object Definiteness and Voice*

- a. Na-alli-ang i [_S iKaco'] [_O itim bunga] [_D iCicci'].
 3ERG-buy-APPL 3ABS NAME that flower NAME
 ‘Kacho’ bought that flower for Chichi’.
- b. Na-alli i [_S iKaco'] [_O itim bunga].
 3ERG-buy 3ABS NAME that flower
 ‘Kacho’ bought that flower.’

The order VSOD is also possible in the Mandarin version of a construction called the “Agent Voice” in many Western Austronesian languages (Kroeger 1993a). In Mandarin, this voice is formed by replacing the ergative prefixes on the V with the affix *may-*. This voice has three properties: it requires the S to be absolutive; it requires the O and D to be indefinite; and it allows the O and D to be implicit. Following the ergative tradition of analysis of Western Austronesian syntax, I will refer to this voice as an antipassive (De Guzman 1988; Aldridge 2004; Paul and Travis 2006). The following examples show clauses where the verb takes this voice: the first shows the order VSOX and the second the order VSODX.

(9) *VSOD Order: The Antipassive (Agent Voice)*

- a. Mam-baca i [_S iKaco'] [_O buku] dio.
 ANTIP-read 3ABS NAME book there
 ‘Kacho’ is reading a book there.’
- b. Mam-be-ngang i [_S iKaco'] [_O buku] [_D passikola] dio.
 ANTIP-give-APPL 3ABS NAME book schoolkid there
 ‘Kacho’ is giving books to schoolchildren there.’

There is also a context in Mandarin where the order VSOD is required. When postverbal arguments appear in the space before locative and temporal adjuncts, they are systematically forced to surface in the order S-O-D. This restriction holds no matter the definiteness of the object or the voice of the verb, and the following examples illustrate: NP-NP-NP strings in this position must always be interpreted as S-O-D when the verb is ditransitive (10a) and when it is antipassive (10b).

(10) *V-NP-NP-NP-X* → *Strictly VSODX*

- a. Na-alli-ang i [s iKaco'] [o iGary] [D iCicci'] dio.
 3ERG-buy-APPL 3ABS NAME NAME NAME there
 ONLY VSODX: 'Kacho' bought Gary (a local cat name) for Chichi' there.'
- b. Mak-kiring-ang i [s passikola] [o dottor] [D guru'] dio.
 ANTIP-send-APPL 3ABS student doctor teacher there
 ONLY VSODX: 'The student sent a doctor to a teacher there.'

This initial picture is complicated by the following fact: in Mandar, it is always possible for definite arguments to surface in positions that follow locative and temporal adjuncts. The following examples illustrate, building from the ditransitive VSODX clause in (10a). The S must always be definite in Mandar, and as a result, this argument can always take positions that follow locative and temporal adjuncts to the VP (11a). When the O is indefinite, it cannot surface in positions of this type (11b). But when the O is definite, it can freely do so (11c).

(11) *Definite Arguments* → *Optionally Follow Adjuncts to the VP*

- a. Na-alli-ang i [o iGary] [D iCicci'] dio [s iKaco'].
 3ERG-buy-APPL 3ABS NAME NAME there NAME
 'Kacho' bought Gary for Chichi' there.'
- b. *Na-alli-ang i [s iKaco'] [D iCicci'] dio [o posa].
 3ERG-buy-APPL 3ABS NAME NAME there cat
 INTENDED: 'Kacho' bought a cat for Chichi' there.'
- c. Na-alli-ang i [s iKaco'] [D iCicci'] dio [o iGary'].
 3ERG-buy-APPL 3ABS NAME NAME there NAME
 'Kacho' bought Gary for Chichi' there.'

Because definite arguments can always surface in multiple positions, the order of arguments in the postverbal domain often seems to be free. In clauses that lack locative and temporal adjuncts, more specifically, V-NP-NP-NP strings typically support multiple readings when the the phonology is not held constant. The following example shows a first corner of this system: written V-NP-NP strings can always be interpreted as either VSO or VOS.

(12) *Surface Ambiguity in Written V-NP-NP Strings*

Ma'-itai i [?? dottor] [?? guru'].
 ANTIP-look 3ABS doctor teacher

VSO: 'The doctor is looking for a teacher.'

VOS: 'The teacher is looking for a doctor.'

This flexibility gives rise to further ambiguity in strings of the shape V-NP-NP-NP. In ditransitive clauses of this shape, the language allows the six interpretations VSOD, VSDO, VOSD, VDSO, VODS, and VDOS when the phonology is not fixed.

(13) *More Surface Ambiguity*

- a. Na-alli-ang i [?? iKaco'] [?? iGary] [?? iCicci'].
 3ERG-buy-APPL 3ABS NAME NAME NAME
- b. VSOD: 'Kacho' bought Gary for Chichi'.
 VSDO: 'Kacho' bought Chichi' for Gary'.
 VOSD: 'Gary bought Kacho' for Chichi'.
 VDSO: 'Gary bought Chichi' for Kacho'.
 VODS: 'Chichi' bought Kacho' for Gary'.
 VDOS: 'Chichi' bought Gary for Kacho'.

Whatever syntactic mechanisms lie beneath these alternations, it is important to note at the outset that they can be distinguished from other processes that influence word order and correlate with clear pragmatic effects. The first of these is extraposition, which places given arguments in a right-peripheral position where they form separate intonational phrases (14a) (*cf.* Antinucci and Cinque 1977; Aissen 1992). The second is a type of focus movement that draws contrastive foci to the right edge of the clause, forcing its targets to receive an exceptional form of word-final stress (marked with a grave accent, (14b)). In the investigation that follows, we will set these two processes aside to focus on the pragmatically unmarked alternations above.

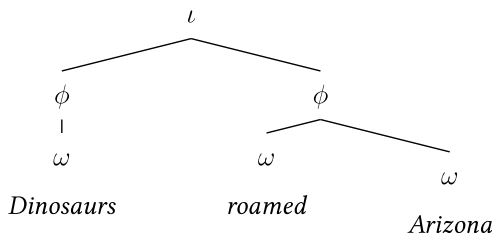
(14) *Extraposition and Rightward Focus-Movement*

- a. Na-alli-ang i [o iGary] [D iCicci'] dio, [s iKaco'].
 3ERG-buy-APPL 3ABS NAME NAME there NAME
 'He bought Gary for Chichi' there, Kacho'.
- b. Na-alli-ang i [o iGary] [D iCicci'] dio [s iKacò'].
 3ERG-buy-APPL 3ABS NAME NAME there NAME
 'KACHO' bought Gary for Chichi' there (... , not Ali).'

2.2 Phonological constituency tests

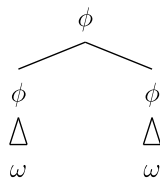
To study the syntax of these alternations, we will turn to the facts of phonology. At the metrical level, segments are organized into syllables and syllables are parsed into feet, yielding a hierarchical constituent structure that is built from abstract phonological constituents (Lieberman and Prince 1977). The theory of the Prosodic Hierarchy (Selkirk 1978, 1984, 1986; Nespor and Vogel 1986) holds that the same type of hierarchical phonological constituent structure extends above this level: feet are organized into prosodic words and prosodic words are organized into phonological phrases.

Following Itô and Mester (2007), I will assume that this type of suprametrical prosodic structure is systematically built from three universal abstract phonological building blocks: the prosodic word (ω), phonological phrase (ϕ), and intonational phrase (ι). An example of this structure is shown in diagram (15), which reflects the parse of the clause *Dinosaurs roamed Arizona*. In that clause, the subject, verb, and object are each mapped to ω s; the verb and object are then parsed into a single ϕ .

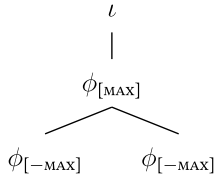
(15) *Phonological Organization above the Word*

As we investigate the prosody of the Mandarin VP, we will build on three results that are now widely accepted in the mainstream literature on prosody. The first is the understanding that the relationship between syntax and prosody is indirect (the Indirect Reference Hypothesis: Nespor and Vogel 1986). The constituents that provide the domains for phonological events and phonotactic restrictions in the phrasal phonology must be phonological constituents, like the ω , ϕ , and ι , rather than syntactic constituents. In the same vein, these constituents are organized into a hierarchical structure that mirrors—but does not exactly match—that of the syntax.

The second foundational hypothesis surrounds *recursion*. Prosodic constituents must be able to dominate elements of the same type on the prosodic hierarchy (16). This result has been established decisively at the level of the ϕ , which shows patterns of embedding that cannot be captured by stipulating additional strictly-layered constituents (Ladd 1986; Wagner 2005, 2010; Itô and Mester 2007, 2012, 2013; Elfner 2012, 2015; see also Itô and Mester 1992; Martínez-Paricio and Kager 2015; Bennett 2018). This fact has led to the generalized adoption of Weak Layering Theory (Itô and Mester 1992), which allows for prosodic recursion, unbalanced sisterhood (ω s can be sisters to feet), and level-skipping succession (ω s can directly dominate syllables, so long as they have heads).

(16) *Prosodic Recursion*

Our final guiding hypothesis is the understanding that the phonology can recognize and exploit relational distinctions between phonological constituents that are recursively nested (Elfner 2012, 2015; Itô and Mester 2012, 2013; Elordieta 2015). More specifically, I will assume that phonotactic restrictions and prosodic events can be keyed to the edges of ϕ s that are directly dominated by an ι , thus failing to occur at the edges of ϕ s that are nested within higher ϕ s. Diagram (17) illustrates this type of unembedded ϕ ; in the relational terms of Itô and Mester (2012, 2013), it is the maximal ϕ . I will denote this type of ϕ as the $\phi_{[\text{MAX}]}$.

(17) *The Maximal ϕ* 

2.2.1 The building blocks

We can now introduce language-internal tests for the building blocks of prosodic organization. At the metrical level in Mandarin, a single instance of stress falls on every constituent that corresponds to a head at the base of an extended projection (Grimshaw 1991): N^0 , V^0 , ADJ^0 , and ADV^0 (including many adverbs that surface in second-position). It is absent from heads that sit higher on the functional spine, such as P^0 , DEM^0 , and C^0 . This stress surfaces once per word and falls on the penultimate syllable (Pelenkahu et al. 1983), and its most salient phonetic correlate is a low tone (co-occurring with increases in amplitude and length). The following example shows its distribution with the format that I will use below: the top line shows an abstract surface form of the utterance and the second shows its underlying form.⁵

(18) *Penultimate Stress*

óro oróaŋ	oroánna
oro oro-aŋ	oro-aŋ-na
sit	sit-NOMINALIZER seat-NOMINALIZER-3GEN

‘to sit, a seat, her seat’

I propose that this stress marks the right edge of the prosodic word (ω). The distribution of stress thus suggests that N^0 , V^0 , ADJ^0 , and ADV^0 always form ω s in Mandarin, as elsewhere (McCarthy and Prince 1993; Selkirk 1996, 2009; Truckenbrodt 1999).

Above the level of the ω , there is another prosodic constituent that we can easily detect. Most types of XPs in Mandarin are followed by a single final high tone (H). The following example shows its distribution: it appears after sequences of X^0 s and X^0 -level adjuncts, like $ADV-V$ and $N-ADJ$ (19). I will assume that this H falls at the right edge of a constituent above the ω : the phonological phrase (ϕ).

(19) *The Phonological Phrase*

máne wémme ^H i	iti	āndʒóro káiāŋ ^H ó ^H
mane bemme i	itiŋ	andʒoro káiāŋ ^H ó ^H
just fall	3ABS	that coconut big there

‘That big coconut there just fell.’

⁵Regular secondary stress has not been described in other languages of the subfamily: Mithun and Basri (1986) do not mention secondary stress in Selayarese, while Friberg and Friberg (1991) and Jukes (2006) argue that it is absent in Konjo and Makassarese. I will assume that it is absent in Mandarin as well.

2.2.2 The maximal phonological phrase

The next-highest level of prosodic organization can be seen from the distribution of specific restrictions in the phrasal phonology. The following section presents four.⁶ First, Mandarin has a process of Coalescence that reduces /ai ae/ to [e] and /au ao / to [o]. The following example shows how this process operates in three-word VSX clauses under default prosodic conditions. When the verb ends in a vowel sequence it must show Coalescence (*lnaitail* ‘seek’ → [néte]). But when the S ends in a final vowel sequence, it cannot (*lbalaol* ‘mouse’ → [wálo]). The same restriction holds over the X (*/di bangae/* → [di wángge]).

(20) Coalescence: The First Asymmetry

néte ^H	i	waláo ^H	di	wángáe ^H
na-itai	i	balao	di	bangae
3ERG-look for	3ABS	mouse	in	PLACE

‘The mouse looked for it in Banggae.’

Second, Mandarin has a process of Gliding that reduces prevocalic /i u/ to [j w] in the same positions. Under default prosodic conditions in three-word clauses of the shape VSX, this process applies at the right edge of the V, but not at the right edges of the S or X. The following example illustrates: the V */napapia/* ‘make’ must reduce to [napápja], but the S */imaria/* ‘NAME’ cannot reduce to [imárja] (21). The same restriction extends to the final X (*/di lamasariaŋ/* → [di lamasárjaŋ]).

(21) Gliding: A Second Asymmetry

napápja ^H	i	imaría ^H	di	lamasaríjaŋ ^H
na-papia	i	imaria	di	lamasariaŋ
3ERG-make	3ABS	NAME	in	PLACE

‘Maria made it in Lamasariang.’

Third, the same restrictions emerge around a process of Glottal Coda Deletion. In Mandarin, intervocalic /ʔ/ is always syllabified as a coda and intervocalic /ŋ/ is syllabified as a coda word-finally (on the glottal status of /ŋ/: De Lacy 2006). The following example shows how these segments are realized intervocalically in three-word VSX clauses under default prosodic conditions. At the right edge of the V, Glottal Coda Deletion must apply, reducing */nakaraʔus/* ‘scratch’ to [nakáros]. But at the right edge of the S, this process is blocked: */iripaʔi/* ‘NAME’ cannot reduce to [irípe] (22). The same restriction extends to the X (*/di rebataʔa/* → [di rewáta]).

⁶There are more phonological processes in Mandarin that are blocked at the edges of the same higher prosodic constituent (the $\phi_{[MAX]}$), including more complex processes triggered by the absolutive agreement clitics. For ease of exposition, these processes will not be reflected in the examples below.

(22) *Glottal Coda Deletion: A Third Asymmetry*

nakáros^H i iripáʔi^H di rewatáʔa^H
 na-karaʔus i iripaʔi di rebataʔa
 3ERG-scratch 3ABS NAME in PLACE

‘Ripa’i scratched it in Rebata’a.’

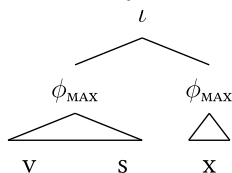
Fourth, the same patterns emerge around a process of Voiced Obstruent Lenition. At the junctures that require Coalescence, Gliding, and Glottal Coda Deletion at the right edge of one constituent, intervocalic /b d d̥ ʒ g/ must lenite to [w ɹ j ɣ] at the left edge of the next. The following example illustrates in a three-word VSX clause. Under default prosodic conditions, the right edge of the V must show Coalescence, Gliding, and Glottal Coda Deletion in this context. At the same time, the left edge of the S must show Voiced Obstruent Lenition when the preceding segment is a vowel. As a result, the S /do gurul/ ‘that teacher’ must be realized as [ɹo yúru] (23). As Coalescence, Gliding, and Glottal Coda Deletion are blocked at the right edge of the S, Voiced Obstruent Lenition is also blocked at the left edge of the X (/di genaʔ/ ‘earlier’ → [ɹi yénaʔ]).

(23) *Voiced Obstruent Lenition: A Fourth Asymmetry*

nawátʃa^H i ɹo yúru^H di yénaʔ^H ó^H
 na-batʃa i do guru di genaʔ o
 3ERG-read 3ABS that teacher in earlier there

‘That teacher there read it earlier.’

These restrictions suggest the pattern of constituency in diagram (24): the VS string forms a domain that allows Coalescence, Gliding, Glottal Coda Deletion, and Voiced Obstruent Lenition internally. The right edge of this constituent blocks the first three processes, banning final Coalescence, Gliding, and Glottal Coda Deletion in a two-syllable window at the right edges of the S and X. The left edge of this constituent blocks the last process, banning initial Voiced Obstruent Lenition in the X. As these restrictions cannot be keyed to syntactic constituents, we must identify the relevant domain as a prosodic constituent that falls between the minimal ϕ and the intonational phrase (ι). I propose that this constituent is the maximal phonological phrase, or ϕ_{MAX} . On this analysis, the minimal ϕ s that contain the V and the S are parsed into a higher ϕ , yielding a pattern of recursive prosodic constituency. This higher ϕ is directly dominated by the ι , and as a result, it acquires the relational property of maximality.

(24) *The Parse of VSX*

On this analysis, we can understand the restrictions on Voiced Obstruent Lenition, Coalescence, Gliding, and Glottal Coda Deletion to provide diagnostics for the left and right edges of the maximal ϕ . The following table integrates this result with our earlier observations on stress and the H tone to summarize the diagnostics that will allow us to study prosodic constituency below.⁷

(25) *Prosodic Constituents and Diagnostics*

CONSTITUENT	LEFT EDGE	RIGHT EDGE
ω		Stress
ϕ		H tone
$\phi_{[\text{MAX}]}$	✗Voiced Obstruent Lenition	✗Coalescence ✗Gliding ✗Glottal Coda Deletion

3 The structure of the VSOD string

Suprametrical prosodic constituent structure is roughly grounded in the syntax beneath it. The phonological phrase (ϕ), for instance, is widely understood to be built in response to an interface pressure to preserve the distribution of XPs (Selkirk 1984, 1986; Nespor and Vogel 1986; Selkirk and Tateishi 1988; Inkelas 1989; Selkirk and Shen 1990; Selkirk and Tateishi 1991; Inkelas and Zec 1995; Selkirk 1995, 2009; Elfner 2012, 2015). Following Itô and Mester (2019a), I will assume that this requirement is one of existential correspondence, demanding that every input XP be placed in correspondence with an output ϕ (26a). The alignment between these XPs and their corresponding ϕ s is then forced by the PROSODY-TO-SYNTAX MATCH constraint in (26b).

(26) *The Isomorphism Constraints*

- MAX-XP: Let SX be an input syntactic representation and PR its corresponding output representation. Assign one violation for every XP in SX which does not correspond to a ϕ in PR.

⁷The ϕ_{MAX} can also be distinguished from the intonational phrase (ι), which is built around matrix clauses, parentheticals, preposed adjunct CPs, left- and right-topics, and extraposed constituents. The right edge of the ι blocks Nasal Assimilation: coda /ŋ/ assimilates to all following segments and denasalizes before all non-nasals but /b d dʒ g/ (Pater 1999), across ω - and ϕ -boundaries but not across the ι edge. Example (i) shows it cannot apply between a topic and v but must apply between an s and an x.

(i) *Nasal Assimilation: A Diagnostic for the ι*

$\{ \iota \{ \phi_{[\text{MAX}]} \} \}$	$\{ \iota \{ \phi_{[\text{MAX}]} \} \}$	$\{ \iota \{ \phi_{[\text{MAX}]} \} \}$	$\{ \iota \{ \phi_{[\text{MAX}]} \} \}$
itim búlaŋ ^H ,	póle ^H i	irámas ^H	sola wenéna ^H
itiŋ bulaŋ	pole i	iramaŋ	sola baine-na
that month	come 3ABS	NAME	with wife-3GEN

‘That month, Ramang came with his wife.’

- b. $\text{MATCH}(\phi, \text{XP})$: Let SX be an input syntactic representation and PR its corresponding output representation. Assign one violation for every ϕ in PR that has a corresponding XP in SX such that the left and right edges of the ϕ are not aligned with the left and right edges of the XP .

These constraints work together to parse strings into prosodic structures that roughly reflect the syntax beneath them. For instance, the mapping from maximal XPs to ϕ s forces DPs and PPs to form ϕ s in Mandarin, as we have already seen. But the mapping constraints have also been argued to force the recursive mapping of functional projections to ϕ s as well (Elfner 2012, 2015; Itô and Mester 2019a). As a result, we might take the mapping of the VSX string to reveal a fact about the syntax: as the v and S are typically parsed into ϕ s (27a), they may form an XP in the syntax (27b).

(27) *Extrapolating XPs from ϕ s*



The task of this section is to build from this starting case to a syntactic result. We will first build a prosodic argument that the VSOD string must always form a surface constituent when it appears before an X , and we will then develop an account of the syntax that underlies the Mandarin strings of this shape.

3.1 The prosody of the VSOD string

We can begin our investigation with a brief survey of the prosody of adjunction. Under default prosodic conditions, we have seen that the final X in a VSX string forms its own maximal ϕ . In clauses that contain multiple X s, we can now observe that the same parse is extended to each one. Example (28) illustrates: when a VS string is followed by the VP -level adjuncts *dio* ‘there,’ *dionging* ‘yesterday’ and *o* ‘there,’ Voiced Obstruent Lenition and Glottal Coda Deletion are blocked between every X . These restrictions reveal the parse $\{\phi_{[\text{MAX}]} \text{VS}\} \{\phi_{[\text{MAX}]} \text{X}\} \{\phi_{[\text{MAX}]} \text{X}\} \{\phi_{[\text{MAX}]} \text{X}\}$.

(28) *VSX Clauses: Each Adjunct = ϕ_{MAX}*

$\{\phi_{[\text{MAX}]}$				$\{\phi_{[\text{MAX}]}\}$	$\{\phi_{[\text{MAX}]}\}$	$\{\phi_{[\text{MAX}]}\}$
nasáka ^H	i	ɿo	waláo ^H	díó ^H	djóngjɿ ^H	ó ^H
na-saka	i	do	balao	dio	diongjɿ	o
3ERG-catch	3ABS	that	mouse	there	yesterday	there

‘That mouse caught it there yesterday.’

The same parse extends to all final adjuncts to the VP . This fact can be seen from the behavior of PPs. When PPs are selected by the v in Mandarin, they are parsed into maximal ϕ s with the VS string under default prosodic conditions in three-word clauses of the order VS-PP . In example (29a), for instance, the comitative PP *sola iMina* ‘with Mina’ is selected by the v *siala* ‘marry’ and the parse is

$\{\phi_{[MAX]} \text{ VS PP} \}$. When PPs are adjoined to the VP, in contrast, they are forced to form maximal ϕ s alone. Thus in example (29b), where the same PP appears after the V *lambda* ‘go,’ the parse is $\{\phi_{[MAX]} \text{ VS} \} \{\phi_{[MAX]} \text{ PP} \}$.

(29) *The Parse of PPs depends on Argument-Adjunct Status*

- a. $\{\phi_{[MAX]} \quad \quad \quad \}$ $\{\phi_{[MAX]} \quad \quad \}$
- | | | | | |
|--------------------|------|--------------------|-------------------------|----------------------|
| sjála ^H | i | irípe ^H | sola imína ^H | djónjij ^H |
| si-ala | i | iripa?i | sola imina | dionjij |
| COM-take | 3ABS | NAME | with NAME | yesterday |
- ‘Ripa’i took with Mina yesterday (= they got married).’
- b. $\{\phi_{[MAX]} \quad \quad \quad \}$ $\{\phi_{[MAX]} \quad \quad \}$ $\{\phi_{[MAX]} \quad \quad \}$
- | | | | | |
|--------------------|------|----------------------|-------------------------|----------------------|
| lámba ^H | i | iripá?i ^H | sola imína ^H | djónjij ^H |
| lambda | i | iripa?i | sola imina | dionjij |
| go | 3ABS | NAME | with NAME | yesterday |
- ‘Ripa’i went with Mina yesterday.’

These restrictions reveal a generalization that recalls the results of much earlier work on the prosody of adjunction (Truckenbrodt 1999; also Cinque 1993). Mandarin requires right-adjuncts to the VP to be parsed into separate phrase-level prosodic domain from the rest of the extended VP, much like Xiamen (Chen 1987) and Tohono O’odham (Hale and Selkirk 1987). Mandarin also requires every X to form its own maximal ϕ , banning multiple Xs from forming maximal ϕ s together. In the same vein, the presence of an X has no effect on the parse of the preceding VSOD string. We thus begin our investigation from the starting generalization in (30).

(30) *The Generalization on Adjunction*

The presence of VP-level adjuncts does not influence the parse of the VP.

Every VP-level adjunct forms a ϕ_{MAX} of its own under default prosody.

We can now turn to the prosodic constituency of the VSODX string. To begin, we will focus on the default prosody of clauses in which the sequence of the verb and its arguments forms exactly three ω s (reserving the study of four- and five- ω clauses for Sect. 5). The following example introduces our first generalization in this domain: when the verb is transitive, VSOX strings must be parsed $\{\phi_{[MAX]} \text{ VSO} \} \{\phi_{[MAX]} \text{ X} \}$. In other words, before an X, a three- ω transitive VSO string must form a maximal ϕ .

(31) *VSO = ϕ_{MAX} (Transitive)*

- $\{\phi_{[MAX]} \quad \quad \quad \}$ $\{\phi_{[MAX]} \quad \quad \}$ $\{\phi_{[MAX]} \quad \quad \}$
- | | | | | | |
|-------------------|------|--------------------|----------------------|-------------------|----------------|
| néte ^H | i | irípe ^H | je wúku ^H | díni ^H | é ^H |
| na-itai | i | iripa?i | de buku | dini | e |
| 3ERG-look for | 3ABS | NAME | this book | here | here |

‘Ripa’i is looking for this book here.’

When the verb is ditransitive, VSOX strings receive the same parse. The following examples illustrate, varying the definiteness of the O: whether indefinite (32a) or definite (32b), the VSOX string must be parsed $\{\phi_{[MAX]} \text{ VSO} \} \{\phi_{[MAX]} \text{ X} \}$.

(32) $VSO = \phi_{MAX}$ (Ditransitive)

- a. $\{\phi_{[MAX]}$ $\}$ $\{\phi_{[MAX]}$ $\}$
 nakiríŋa^H **ã** irípe^H wúku^H djóŋŋ^H
 na-kiriŋ-aŋ a? iripa?i buku dionŋŋ
 3ERG-send-APPL 1ABS NAME book yesterday
 ‘Ripa’i sent me a book yesterday.’
- b. $\{\phi_{[MAX]}$ $\}$ $\{\phi_{[MAX]}$ $\}$ $\{\phi_{[MAX]}$ $\}$
 nakiríŋa^H **ã** irípe^H **ɛ** wúku^H djóŋŋ^H **é^H**
 na-kiriŋ-aŋ a? iripa?i de buku dionŋŋ e
 3ERG-send-APPL 1ABS NAME this book yesterday here
 ‘Ripa’i sent me this book here yesterday.’

The same parse holds over three- ω strings of the orders VSD and VOD. Example (33a) shows that VSDX strings must be parsed $\{\phi_{[MAX]}$ VSD $\}$ $\{\phi_{[MAX]}$ X $\}$; example (33b) shows that VODX strings must be parsed $\{\phi_{[MAX]}$ VOD $\}$ $\{\phi_{[MAX]}$ X $\}$.

(33) $VSD / VOD = \phi_{MAX}$ (Ditransitive)

- a. $\{\phi_{[MAX]}$ $\}$ $\{\phi_{[MAX]}$ $\}$
 najolóa^H **ĩ** irípe^H yúru^H djóŋŋ^H
 na-d̥ɔlo-aŋ i iripa?i guru dionŋŋ
 3ERG-point-APPL 3ABS NAME teacher yesterday
 ‘Ripa’i showed it to the teacher yesterday.’
- b. $\{\phi_{[MAX]}$ $\}$ $\{\phi_{[MAX]}$ $\}$
 ujolóa^H **ĩ** wálo^H yúru^H djóŋŋ^H
 u-d̥ɔlo-aŋ i balao guru dionŋŋ
 1ERG-point-APPL 3ABS mouse teacher yesterday
 ‘I showed the teacher a mouse yesterday.’

In a specific phonological context, we can see that strings of the orders VSO, VSD, and VOD are also parsed into maximal phrases when the verb is antipassive. In Mandarin, the antipassive prefix *maŋ-* carries an exceptional instance of word-level stress (34a). Before vowel-initial roots, this prefix retains its stress and is realized as [má?-] when it is explicitly protected by the phonology of focus (for instance, when speakers are asked to provide citation forms of vowel-initial verbs in the antipassive voice). But under default prosodic conditions, the nucleus of this prefix is erased before vowel-initial roots by the joint work of Glottal Coda Deletion and Coalescence. The prefix thus loses its word-level stress and takes the form *m-* (34b).

(34) *The Phonology of the Antipassive Prefix*

- a. **má?**ánde, **máttú**nu, **mámbá**wa
 maŋ-ande maŋ-tunu maŋ-baba
 ANTIP-eat ANTIP-roast ANTIP-bring
 ‘Eat, roast, bring.’

- b. $\{\phi_{[MAX]}$ $\}$ $\{\phi_{[MAX]}$ $\}$
mánde^H i **jóŋa^H** díó^H
maŋ-ande i dʒoŋa dio
ANTIP-eat 3ABS deer there

‘The deer are eating there.’

The familiar parses of VSO, VSD, and VOD strings emerge in antipassive clauses when the antipassive prefix loses stress in this way. The string VSOX is parsed $\{\phi_{[MAX]}$ VSO $\}$ $\{\phi_{[MAX]}$ X $\}$ (35a), the string VSDX is parsed $\{\phi_{[MAX]}$ VSD $\}$ $\{\phi_{[MAX]}$ X $\}$ (35b), and the string VODX is parsed $\{\phi_{[MAX]}$ VOD $\}$ $\{\phi_{[MAX]}$ X $\}$ (35c).

(35) VSO / VSD / VOD: = ϕ_{MAX} (Antipassive)

- a. $\{\phi_{[MAX]}$ $\}$ $\{\phi_{[MAX]}$ $\}$
mánde^H i **jóŋa^H** **jólen^H** díó^H
maŋ-ande i dʒoŋa dʒoleŋ dio
ANTIP-eat 3ABS deer guava there

‘The deer are eating guavas there.’

- b. $\{\phi_{[MAX]}$ $\}$ $\{\phi_{[MAX]}$ $\}$
mállja^H **ĩ** **irípe^H** **ýúru^H** djóŋiŋ^H
maŋ-alli-aŋ i iripaʔi guru dionŋiŋ
ANTIP-buy-APPL 3ABS NAME teacher yesterday

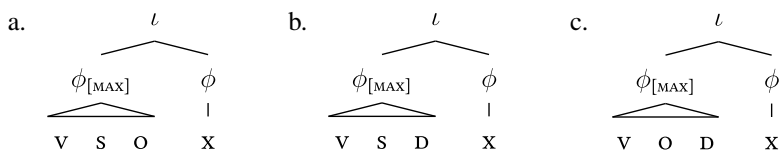
‘Ripa’i was buying stuff for teachers yesterday.’

- c. $\{\phi_{[MAX]}$ $\}$ $\{\phi_{[MAX]}$ $\}$
mállja^H **ĩ** **wúku^H** **ýúru^H** djóŋiŋ^H
maŋ-alli-aŋ i buku guru dionŋiŋ
ANTIP-buy-APPL 3ABS book teacher yesterday

‘He was buying books for teachers yesterday.’

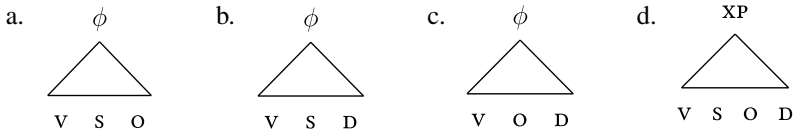
We thus arrive at a stable prosodic generalization: in the space before an X, where the word order is strictly VSOD, three-word substrings of the order VSOD are always parsed into single maximal ϕ s. The following diagrams show this starting result.

(36) Stable Prosodic Constituency



3.2 Convergent diagnostics

As three-word VSO, VSD, and VOD strings form maximal ϕ s before Xs under default prosodic conditions, I propose that these strings always form XPs when they precede an X. Extending this conclusion to four-word strings of the order VSOD, we arrive at the result in (37d): the VSOD string can form a syntactic constituent.

(37) *Prosodic Constituency to Syntax*

We can now test this analysis against independent diagnostics from the syntax. The first of these involves displacement. There is a construction in Mandarin in which the control verb *pogau* ‘do’ selects a non-finite predicate that raises into the left periphery (38). I will refer to this process as Predicate Clefting.

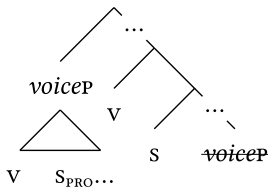
(38) *Predicate Clefting*

[_{XP} **Mam-baca** tappa’ PRO] na-pogau’ iBa’d� [~~mam-baca~~ tappa’ PRO] .
 ANTIP-read only 3ERG-do NAME read only

‘All that Ba’d� does is read.’

Sikki et al. (1987, 783)

I assume that the constituent that moves under Predicate Clefting is the XP in (37d). I will label this constituent as the *voiceP* and take it to move to the left-peripheral landing site of WH-movement in Mandarin (Brodtkin 2020), as in (39).

(39) *The Syntax of Predicate Clefting*

This footwork sets us up with a prediction about VSOD strings. As these strings form maximal ϕ s before Xs, we should predict that they should also form constituents for Predicate Clefting (or at least VOD strings should, as the S must be null and controlled in the *voiceP*s fronted by this process). This prediction is correct. No matter the voice of the V or the definiteness of the O, VOD strings can always be clefted (40).

(40) *Stable Constituency for VOD*

- a. [_{voiceP} Mam-bengang buku passikola] na-pogau’ iAli ____.
 ANTIP-give book schoolkid 3ERG-do NAME
 ‘What Ali does is give students books.’
- b. [_{voiceP} Na-alli-ang lo’diang iCicci’] na-pogau’ iKaco’ ____.
 3ERG-buy-APPL ring NAME 3ERG-do NAME
 ‘What Kacho’ does is buy Chichi’ rings.’

The facts of Predicate Clefting thus reconfirm the results of our prosodic investigation: VSOD strings must form tight constituents in the syntax. We can now reinforce

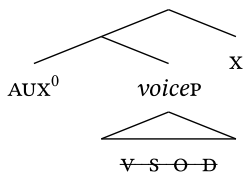
this result with a test from the domain of ellipsis. Mandar has a type of ellipsis that targets the complements of aspectual auxiliaries and the negator (41). This process has much in common with the VP-ellipsis (VPE) of English and Indonesian (Fortin 2007): it can license sloppy readings, occur in embedded clauses, and involve antecedents in embedded clauses (Hankamer 1979; Johnson 2009).

(41) *VP-Ellipsis in Mandar*

- a. U-sanga ndangi **rua** [_{voiceP} na-pikkiri iAli irama],
1ERG-think not.3ABS have 3ERG-think.of NAME prosody
'I thought that Ali had never thought about prosody.'
- b. Yari tiwikke' a' apa' **rua** i [_{XP} ~~na-pikkiri~~ iAli irama].
so shocked 1ABS because have 3ABS thought.of NAME prosody
'... So I'm shocked because he has ~~thought about prosody~~.'

I will assume that VP-ellipsis suppresses a constituent that sits just beneath the licensing auxiliary: plausibly, the *voiceP* (42). This analysis sets up a prediction. If the VSOD string always forms a syntactic constituent before an X, then VP-ellipsis should always suppress the full VSOD string. In other words, this ellipsis should render it impossible for the S, O, and D to surface between a licensing auxiliary and an X.

(42) *The Constituency Prediction: Ellipsis*



This prediction is correct. When VPE suppresses an antipassive V, it bans the appearance of an S between the auxiliary and an adjunct. This is shown in example (43b), which shows that neither the S nor a coreferent epithet can appear there.

(43) *The Constituency of the VS String*

- a. Usanga di ruambongi ndangi rua maccoro iKaco'.
'I used to think that Kacho' had never stolen anything.'
- b. Mane u-issang i [_{CP} mua' [_{PERFP} rua i...
just now 1ERG-know 3ABS that have 3ABS
[_{voiceP} ~~mae-coro~~ (*iKaco' / *do asu)] di duambongi]].
ANTIP-steal NAME that dog in the past
'Now I know that (*Kacho'/*that jerk) has ~~stolen~~ in the past.'

When VPE suppresses a ditransitive V in a VSODX clause, the facts are the same: all material must be silenced between the licensing auxiliary and a postverbal adjunct. As a result, epithets that refer to the S, the O, or the D cannot appear in this space (44).

(44) *Stable Constituency for VSOD*

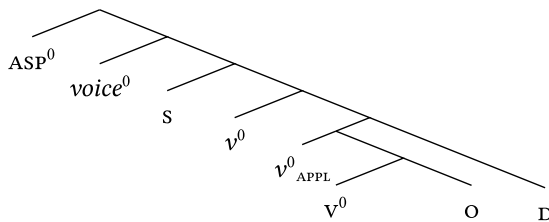
- a. Usanga ndangi rua nawengang iKaco' bulawang gamallo kotta'na.
'I thought that Kacho' had never given fake gold to his girlfriend.'

- b. Mane u-issang i mua' rua i...
 just now 1ERG-know 3ABS that have 3ABS
 [_{voiceP} na-bengane (*do asu) (*do roppong) (*do tokasiasi)] dionging.
 3ERG-give that dog that grass that poor thing recently
 'Now I know (*that jerk) has given (*that junk) (*to the poor girl) recently.'

3.3 The syntax of VSOD

These results allow us to pin down the syntax of the VSOD string. I assume that the syntax beneath the head that licenses VPE has the shape in (45): moving downwards, it contains a *voice*⁰ that hosts the outer morphology of the voice system (Brodtkin 2022b), a *v*⁰ that introduces the s in its specifier (Collins 2005; Merchant 2013), an *appl*⁰ that introduces the D (Pylkkänen 2008), and a *v*⁰ which selects the O. The *voiceP* is the constituent upon which our constituency diagnostics converge: it contains the string VSOD, it hosts VP-level adjuncts at its edge, it is the target of Predicate Clefting and VPE, and it corresponds to the maximal ϕ s that contain the strings VSO, VSD, and VOD (27).⁸

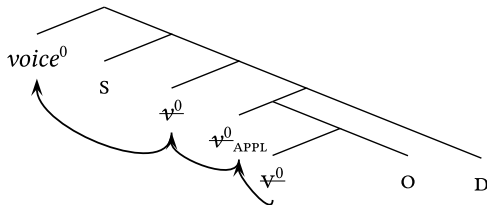
(45) The Syntax of the VSOD String



I assume that the content of the *voiceP* is linearized in the postsyntax (Chomsky 1995), with the order VSOD derived in the following way. The head *appl*⁰ demands that its specifier be linearized to the right, yielding an order where the structurally higher D follows the structurally lower O. The verb, in turn, undergoes a process of postsyntactic head-movement that carries it up its extended projections (Harizanov and Gribanova 2019), amalgamating it with the heads *appl*⁰, *v*⁰, and *voice*⁰ and allowing it to be linearized in a position that precedes the content of the *vP*. This analysis is shown in tree (46).⁹

⁸I will use the labels S, O, and D to represent arguments in the trees below; I will assume that these are NPs when indefinite and DPs when definite. Bare Phrase Structure (Chomsky 1995) creates complexities at the interface; our discussion will sidestep these and use the terminology of X-bar theory (Jackendoff 1977).

⁹The *v* invariably forms a ϕ in the prosody, but this parse is unexpected on Match Theory if it raises into a *X*⁰-position (as it should then form a ω). As a result, this parse must be understood to reflect a mismatch between the syntax and prosody that is driven by a constraint that forces prosodic deviations to satisfy requirements of phonological well-formedness (Selkirk and Elordieta 2010; Itô and Mester 2019b). The natural candidate for such a constraint is STRONGSTART, which mandates various types of phonological strengthening at the left edge of the ι (Selkirk 2011). The operative formulation of this constraint is shown in (ia); it must outrank DEP- ϕ to force the promotion of *v* (see Clemens and Coon 2018; Clemens 2019 for

(46) *The Syntax of VSOD Order in Mandar*


The ensuing analysis captures the word order and constituency of the extended VP with one further addition. Many patterns converge in Mandar to suggest that the argument which triggers absolutive agreement—the S in the antipassive, the O in the transitive, and the D in the ditransitive—always raises to a derived A-position above all other arguments in the clause (Brodkin 2021b, 2022a,b). To provide an example, this movement can be seen in the system of variable binding. The universal quantifier

similar analyses in other verb-initial languages). This effect is shown in tableau (i): STRONGSTART rules out the faithful candidate (a), where v^0 forms a ω , and favors candidate (b), where the V forms a ϕ .

(i) *Deriving the Phrasal Parse of v^0*

- a. STRONGSTART: Assign one violation for every intonational phrase in which the leftmost ω is not left- and right-aligned with a minimal ϕ .

	$[\text{voiceP } v^0 [\text{vP } [\text{DP } S] [\text{VP } _]]]$	STRONG START	DEP- ϕ
a.	$\{ \iota \{ \phi \{ v \{ \phi \{ S \} \} \} \}$	*!	
b.	$\{ \iota \{ \phi \{ \phi \{ v \} \{ \phi \{ S \} \} \} \}$		*

There is direct evidence for this effect in the system of coordination. Mandar has a coordinator which takes the form *na* in ι -medial positions, where it does not carry stress or an H-tone (iia). When a conjunct is extraposed and *na* appears at the left edge of an ι , its parse is changed: there, it shows stress, hosts an H at its right edge, and expands to the disyllabic *anna* (iib). This is similar to the English alternation between 'n and an', which is driven by the need for a new CONJ- ϕ to contain a licit ω .

(ii) *Coordinator Promotion at the Left Edge of the Intonational Phrase*

- a. $\{ \phi[\text{MAX}] \} \{ \phi[\text{MAX}] \}$
- wíta^H i ikátʃoʔ^H na yúru^H di yénaʔ^H
u-ita i ikatʃoʔ na guru di genaʔ
1ERG-see 3ABS NAME and teacher in earlier
- 'I saw Kacho' n'/?²⁷ an' the teacher earlier.'
- b. $\{ \iota \{ \phi[\text{MAX}] \} \} \{ \phi[\text{MAX}] \} \} \{ \iota \{ \phi[\text{MAX}] \} \}$
- wíta^H i ikátʃoʔ^H di yénaʔ^H, á^Hna^H yúru^H
u-ita i ikatʃoʔ di genaʔ na guru
1ERG-see 3ABS NAME in earlier and teacher
- 'I saw Kacho' earlier, an'/*n' the teacher.'

This effect is unique to the coordinator and does not occur with functional heads, like D^0 , P^0 , and C^0 . I assume that this follows from a deeper syntactic split: functional heads like P^0 , D^0 , and C^0 are integrated into extended projections with lexical heads like N^0 and v^0 but the coordinator is not. I thus propose that P^0 , D^0 , and C^0 are parsed into ω s with their complements (they are *internal* clitics; Selkirk 1996) but that the coordinator is not parsed into a ω with its complement in the same way (it is directly dominated by the ϕ , so it is an *external* clitic; Selkirk 1996). Only the coordinator, then, is visible to STRONGSTART.

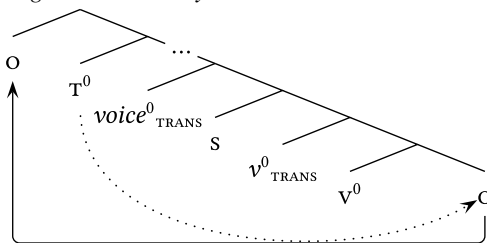
in Mandar, *nasang* ‘every,’ raises to second-position in finite clauses, adjoining to the V in clauses of the order VSOD. When the O is quantified in this way, it cannot bind variables in the preceding S when the O is not absolutive: for instance, in ditransitives (47a). But when O is quantified in this way in a clause where it is absolutive, like a transitive, it gains the ability to bind into the preceding S (47b).

(47) *Variable Binding Shows that the Absolutive Argument Raises Covertly*

- a. Na-kiring-ang nasang a' [s panulis-na] [o buku t].
 3ERG-send-APPL every 1ABS author-3GEN book
 ‘Its_{*i,j} author sent me every book_i.’
- b. Na-itai nasang i [s panulis-na] [o buku t].
 3ERG-look for every 3ABS author-3GEN book
 ‘Its_{i,j} author looked for every book_i.’

Patterns of this sort reveal that Mandar is a High Absolutive language, in the sense of Coon et al. 2014: it requires the absolutive argument to raise to the highest A-position in the clause. This step correlates neatly with the appearance of absolutive agreement in T⁰ (Brodtkin 2021a, 2022b), and it suggests the activity of a classical EPP: in every finite clause, the absolutive argument must raise to SPEC,TP (just as the *pivot* has been argued to raise in other Austronesian languages of the region, setting the terminology of ergativity aside: Schachter 1976; Guilfoyle et al. 1992; Kroeger 1993b; Paul 2000; Richards 2000; Rackowski 2002; Aldridge 2004; Potsdam 2007; Legate 2014; Erlewine 2018; Nie 2019, 2020; Hsieh 2020; Erlewine and Lim 2023). In a transitive clause like (47b), where the facts of variable binding show that the O raises to the highest clause-internal A-position, this conclusion yields the rough syntax in (48) (and see Brodtkin 2022b for detailed discussion).

(48) *High Absolutive Syntax: the Transitive*



The EPP feature localized to this T⁰ must impose no requirements on the spell-out of the chain that it creates. When the absolutive argument raises to SPEC,TP, then, it will never be forced to be pronounced in that position by the output filters that mandate the presence of overt elements in other contexts (Landau 2007b; McFadden and Sundaresan 2018). Instead, the higher copy in SPEC,TP must always be suppressed and the lower copy pronounced: quite possibly in a calculus driven by output-oriented constraints on prosodic organization (Sabbagh 2014) in a case of phonologically-motivated lower-copy spell-out (Bošković 2001, 2002). The result is the rough A-syntax sketched below: different arguments raise across the antipassive, transitive, and ditransitive constructions, but this does not disrupt the emergence of a stable surface constituent that corresponds to the *voiceP*.

(49) *The Rough Syntax of Mandar Voice*

- a. ANTIPASSIVE: [TP S [voiceP V [vP S [applP [VP O] D]]]].
- b. TRANSITIVE: [TP Θ [voiceP V [vP S [VP O]]]].
- c. DITRANSITIVE: [TP Θ [voiceP V [vP S [applP [VP O] D]]]].

4 The internal structure of the VoiceP

The results of our prosodic investigation have thus shown that the VSOD string always forms a surface constituent when followed by adjuncts—but we should also expect to see finer patterns of VP-level constituency than this. Mandar shows many of the asymmetries in binding, coreference, scope, and movement that suggest hierarchical asymmetries between arguments in the *voiceP* (Larson 1988; Pesetsky 1995; Bruening 2001): for instance, applied arguments always seem to C-command internal arguments within the *voiceP*.¹⁰ Through this investigation, then, we have assumed the presence of two specific functional projections in the *voiceP*: a *vP* that contains the string SOD and an *applP* that contains the string OD.

This hypothesis opens up a second line of prosodic inquiry. In Mandar, we cannot detect *voiceP*-internal functional projections with the types of syntactic tests developed in Sect. 3: for instance, either the O or the D can individually raise to a focus position in the left periphery (50a), but the string OD (50b) cannot. This fact holds

¹⁰This fact can be seen in the Mandar construction that Brodtkin (2022b) terms the *Agent Focus*: a voice that morphologically resembles the antipassive, requires the S to be absolutive, and allows the O and D to be definite (it also shows a distinct pattern of ‘accusative agreement’ and contains special morphology in *v*⁰). The following example presents an embedded non-finite clause where the verb surfaces in Agent Focus; in this context, the O and D take their usual order of OD when they appear between the V and an X:

(i) *Agent Focus: Word Order*

U-parriparri'i i [TP **mak**-kiringang i [O iGary] [D iCicci'] dionging].
 IERG-endeavor 3ABS AF-send 3ACC NAME NAME yesterday

ONLY VODX: ‘I endeavored to send Gary to Chichi’ yesterday.’

In this voice, a quantified D can bind into the O (iia), but a quantified O cannot bind into the D (iib).

(ii) *Variable Binding: D > O (Agent Focus)*

- a. U-parriparri'i i [TP **mak**-kiringang nasang i [O buku-nna] [D panulis t].
 IERG-endeavor 3ABS AF-send every 3ACC book-3GEN author
 ‘I endeavored to send every author_i her_{i,j} book.’
- b. U-parriparri'i i [TP **mak**-kiringang nasang i [O buku t] [D panulis-na].
 IERG-endeavor 3ABS AF-send every 3ACC book author-3GEN
 ‘I endeavored to send its_{i,j} author every book_i.’

The same asymmetry emerges in the domain of Condition C (Chomsky 1981; Reinhart 1983): a null pronominal D induces Condition C effects over R-expressions in the O (iia), but a null pronominal O cannot induce Condition C effects over R-expressions in the D (iib)

even though we take it to form a constituent in the syntax—an *applP*—just as the analogous DO string cannot be clefted in English (Takano 2000; Funakoshi 2012).

(50) *Headless Functional Projections: Cannot be Moved*

- a. [_{FP} Buku [_{voiceP} na-bengang i iTalib [_{applP} __ iMina]]].
 book 3ERG-give 3ACC NAME NAME
 ‘It was a book that Talib gave Mina __.’
- b. [_{FP} [_{applP} *Buku iMina] [_{voiceP} na-bengang (i) iTalib _____]].
 book NAME 3ERG-give 3ACC NAME
 INTENDED: ‘*It was Mina a book that Talib gave __.’

The task of this section is thus to build a prosodic case for the existence of headless functional constituents in the extended VP—constituents that are difficult to detect with tests internal to the syntax (for discussion: Takano 2000; Landau 2007a; Funakoshi 2012). Our investigation will take the following path. It has now been established that the TP can be mapped to a ϕ (Elfner 2015; Itô and Mester 2019a). Under the right circumstances, then, it may be possible to detect ϕ s that correspond to headless functional constituents in the *voiceP*, like the *vP* and *applP*. These ϕ s may not always be built and organized into the recursive prosody in (51a), as the mapping constraints interact with other pressures that disturb perfect correspondence and often destroy prosodic constituents that should otherwise be built. But if there are contexts in which we can detect ϕ s that contain the strings SOD and OD, and if we can deduce that such ϕ s are not built for phonological reasons alone, then we can make a prosodic case that the *voiceP* shows the recursive internal syntax in (51b).

(51) *The Prosodic Case for Functional Structure*

- a. POSSIBLE ϕ S: { ϕ_{voiceP} V { ϕ_{vP} S { ϕ_{applP} { ϕ_{VP} O } D } } }
- b. RECURSIVE SYNTAX: [_{voiceP} V [_{vP} S [_{applP} [_{VP} O] D]]]

4.1 The tools for disruption

We can launch this second investigation with a return to the antipassive prefix. When the antipassive prefix *maŋ-* appears before consonant-initial roots in Mandarin, we have seen that it remains syllabic and carries an independent instance of ω -level stress. In connection with this fact, the antipassive prefix is always scanned as a stressed syllable before consonant-initial roots in Mandarin poetry (Jupri Talib, p.c.). All of this

(iii) *Condition C: D > O (Agent Focus)*

- a. U-parriparri'i i [_{TP} **mak**-kiringang i [_O sola-na iNina na iAli] [_D *pro*]].
 1ERG-endeavor 3ABS AF-send 3ACC friend-3GEN NAME and NAME *pro*
 ‘I made an effort to send Nina_i and Ali’s friends to *her**_{i,j}.’
- b. U-parriparri'i i [_{TP} **mak**-kiringang i [_O *pro*] [_D sola-na iNina na iAli]].
 1ERG-endeavor 3ABS AF-send 3ACC *pro* friend-3GEN NAME and NAME
 ‘I made an effort to send *her*_{i,j} to Nina_i and Ali’s friends.’

These facts suggest a hierarchical asymmetry between the O and D in Mandarin: when neither argument is absolutive, the D must C-command the O. See Brodtkin (2022b) for further discussion.

is relevant for its influence on the facts of prosodic phrasing. Up to this point, we have seen that antipassive VSO clauses show the parse $\{\phi_{[MAX]} V SO\}$ when the antipassive prefix is reduced (52a). But in antipassive VSO clauses where the prefix carries stress, the parse shifts to $\{\phi_{[MAX]} V\} \{\phi_{[MAX]} SO\}$ (52b).

(52) *Antipassive Clauses: Phrasing depends on Prefix Reduction*

- a. $\{\phi_{[MAX]}\} \{\phi_{[MAX]}\}$
mánde^H i jóŋa^H jólen^H díó^H
maŋ-ande i dʒoŋa dʒoleŋ dio
ANTIP-eat 3ABS deer guava there
‘The deer are eating guavas there.’
- b. $\{\phi_{[MAX]}\} \{\phi_{[MAX]}\} \{\phi_{[MAX]}\}$
mám bwáí^H i gúru^H wáʔba^H díó^H
maŋ-buai i guru baʔba dio
ANTIP-open 3ABS teacher door there
‘The teacher is opening doors over there.’

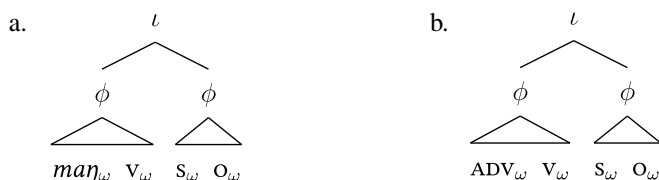
We see the same prosodic shift in all other environments where an additional ω is introduced to the minimal ϕ that contains the verb. The following examples introduce two further cases: adding a X^0 -adjunct like *mane* ‘just’ (53a) or a second-position adverb like *memang* ‘indeed’ (53b) will force the parse $\{\phi_{[MAX]} V\} \{\phi_{[MAX]} SO\}$.

(53) *Prosodic Phrasing is Disrupted by Adjunction to V*

- a. $\{\phi_{[MAX]}\} \{\phi_{[MAX]}\} \{\phi_{[MAX]}\}$
máne nawáí^H i gúru^H ɛ wáʔba^H é^H
mane na-buai i guru de baʔba e
just 3ERG-open 3ABS teacher this door here
‘The teacher just opened this door here.’
- b. $\{\phi_{[MAX]}\} \{\phi_{[MAX]}\} \{\phi_{[MAX]}\}$
náwe mémang^H i gúru^H ɛ wáʔba^H é^H
na-buai memaŋ i guru de baʔba e
3ERG-open indeed 3ABS teacher this door here
‘The teacher opened this door here indeed.’

The following diagrams show this effect: when the minimal ϕ that hosts the verb is augmented with another ω , VSO strings are parsed $\{\phi_{[MAX]} V\} \{\phi_{[MAX]} SO\}$.

(54) *The Disruptions under Ternarity*



These patterns suggest that there is a restriction on the size of the maximal ϕ in Mandarin: it cannot contain more than three ω s. We can derive this effect from the interaction of two constraints on *binarity* (Itô and Mester 1992; Kubozono 1993; Itô and Mester 2013; Ishihara 2014). The constraint MINIMAL BINARITY bans the emergence of maximal ϕ s that only contain one ω (55a). The constraint MAXIMAL BINARITY, in turn, bans the emergence of maximal ϕ s that contain more than two ω s (55b). In the terminology of OT (Prince and Smolensky 2004), we can derive the weight effect by ranking Minimal Binarity over Maximal Binarity. With this ranking, we can force three- ω VSO to be parsed into maximal ϕ s, violating Maximal Binarity to avoid the violations of Minimal Binarity that would emerge around single- ω maximal ϕ s.

(55) *Deriving the Emergence of Ternary Phrases*

- a. MIN-BIN(ϕ_{MAX}): Assign one violation for every ϕ_{MAX} of under two ω s.
- b. MAX-BIN(ϕ_{MAX}): Assign one violation for every ϕ_{MAX} of over two ω s.

c.	$[_{\text{voiceP}} \text{v}^0 [_{\text{VP}} [_{\text{DP}} \text{S}] [_{\text{VP}} [_{\text{NP}} \text{O}]]]]]$	MIN-BIN	MAX-BIN
	 a. $\{\phi_{\text{MAX}} \text{v S O}\}$		*
	b. $\{\phi_{\text{MAX}} \text{v}\} \{\phi_{\text{MAX}} \text{S O}\}$	*!	

With the same ranking, we can also force four- ω strings to be parsed into two maximal ϕ s of two ω s each. The following tableau illustrates the emergence of this parse: in a four-word VSO string where the verbal complex contains two ω s, as in example (52b), the parse must be $\{\phi_{\text{MAX}} \text{v}\} \{\phi_{\text{MAX}} \text{SO}\}$.^{11,12}

(56) *Deriving the Ternarity Effect*

	$[_{\text{voiceP}} \text{man-v}^0 [_{\text{VP}} [_{\text{DP}} \text{S}] [_{\text{VP}} [_{\text{NP}} \text{O}]]]]]$	MIN-BIN	MAX-BIN
	a. $\{\phi_{\text{MAX}} \text{man-v S O}\}$		*
	b. $\{\phi_{\text{MAX}} \text{man-v}\} \{\phi_{\text{MAX}} \text{S O}\}$		

4.2 The exposure effect

This weight constraint opens up a strategy to prosodically split off the *v* from the remainder of the *voiceP*. In this context, the material that follows the *v* will not be

¹¹To capture the absence of this effect in the contexts where the antipassive prefix is reduced, I assume that the construction of the ω that corresponds to the antipassive prefix can be blocked by segmental constraints in the phrasal phonology. This is an expected possibility if prosodic constituents are built in a parallel and global phonological derivation in which the mapping can be disrupted by routine phonological constraints (Prince and Smolensky 2004; Selkirk 2009, 2011)). It is not expected on a cyclic theory in which prosodic structure is built at a stage that precedes the phrasal phonology (e.g., Lee and Selkirk 2023).

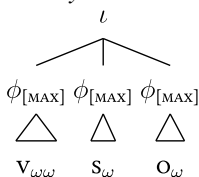
¹²Two other types of affix trigger the same pattern: *p*-initial prefixes which are contained inside the antipassive affixes (*manj-* is underlyingly /N-panj/) and the reduplicant. If these ω -level prefixes are embedded in the ω that corresponds to the *v*, the binarity constraints will need to look into ω s to determine the weight of the ϕ . This would go beyond the versions of leaf-counting Binarity that have been employed in the literature to date (Kalivoda and Bellik 2018; Van Handel 2021). Further work on this topic is required.

parsed into a higher ϕ with the v : instead, it will be directly dominated by the ι . In this context, then, the highest level of ϕ s in the postverbal space will be rendered maximal—and as a result, these previously invisible ϕ s will be rendered visible by our phonotactic tests.

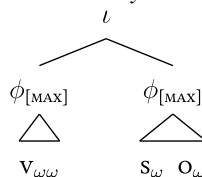
In this context, there are two possibilities that might emerge. The first of these is shown in diagram (57a): if the VSO string is internally flat, containing no internal functional projection that hosts the S and the O, then we should expect the S and O to be parsed into separate maximal ϕ s, with no higher ϕ built around them. The second possibility is shown in diagram (57b): if the VSO string has a recursive internal syntax, containing a functional projection that hosts the S and the O, then the S and the O should be parsed into a single maximal ϕ —one corresponding to a headless vP.

(57) *The Logic of Exposure*

a. *Flat Syntax* $\rightarrow$ *Prosody*



b. *Recursive Syntax* $\rightarrow$ *Prosody*



When we set up the relevant configuration, our diagnostics reveal the parse in (57b): when the v is split off from the SO string, the S and the O are indeed parsed into a single ϕ_{MAX} . The key evidence is shown in example (58): in a transitive VSO clause where the verb hosts a x^0 -level adjunct, Voiced Obstruent Lenition is blocked at the left edge of the S but forced at the left edge of the O. This pattern reveals the existence of an SO- ϕ that is only visible when the verb forms its own maximal ϕ . I will refer to this pattern as the Exposure Effect.

(58) *Weight Manipulations reveal a vP- ϕ*

$\{\phi_{\text{MAX}}\}$			$\{\phi_{\text{MAX}}\}$		$\{\phi_{\text{MAX}}\}$
máne nawáí ^H	i	gúru ^H	ɛ	wáʔba ^H	é ^H
mane na-buai	i	guru	de	baʔba	e
just 3ERG-open	3ABS	teacher	this	door	here

‘The teacher just opened this door.’

Whenever the v is split off in this way, we can see that the same types of ϕ s are built around all two- and three- ω substrings of the order SOD. The following examples present two such cases: ϕ s are built around the substrings SD (59a) and OD (59b).

(59) *The SD and OD Constituents*

$\{\phi_{\text{MAX}}\}$			$\{\phi_{\text{MAX}}\}$		$\{\phi_{\text{MAX}}\}$
máne nannáí ^H	i	do	sanéke ^H	waróʔbo ^H	ó ^H
mane na-annai	i	do	sanaeke	baroʔbo	o
just 3ERG-put	3ABS	that	kid	corn stew	there

‘That kid there just put it in the corn stew.’

- b. $\{\phi_{[MAX]}$ $\}$ $\{\phi_{[MAX]}$ $\}$ $\{\phi_{[MAX]}$ $\}$
- | | | | | | |
|----------|----------------------|------|---------------------|----------------------|----------------|
| nánne | mémaŋ ^H i | do | taláye ^H | waróʔbo ^H | ó ^H |
| na-annai | memaj i | do | talagae | baroʔbo | o |
| 3ERG-put | indeed | 3ABS | that | tomato | corn stew |
| | | | | | there |

‘They did indeed put that tomato there in the corn stew.’

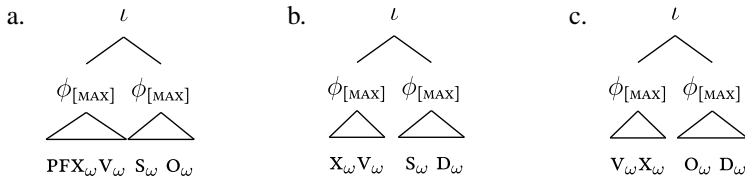
Example (60) shows that the same type of ϕ is built around three- ω SOD strings. There, the minimal ϕ that contains the v is augmented with two further ω s: a x^0 -level adjunct and a second-position adverb. There, Coalescence, Glottal Coda Deletion, and Voiced Obstruent Lenition reveal the parse $\{\phi_{[MAX]} v\} \{\phi_{[MAX]} SOD\}$.

- (60) $\{\phi_{[MAX]}$ $\}$ $\{\phi_{[MAX]}$ $\}$ $\{\phi_{[MAX]}$ $\}$
- | | | | | | | | |
|------|-----------|----------------------|------|-------------------|-------------------|-------------------|----------------|
| máne | nawéŋam | mémaŋ ^H i | do | rípe ^H | wúku ^H | ýúru ^H | ó ^H |
| mane | na-beŋaj | memaj i | do | ripaʔi | buku | guru | o |
| just | 3ERG-give | indeed | 3ABS | that | NAME | book | teacher |
| | | | | | | | there |

‘That Ripa’i there did indeed just give a book to the teacher.’

The following diagrams summarize this result: by manipulating the parse of the verb, we can see that ϕ s can be built around the substrings SO, SD, OD, and SOD.

- (61) *The SOD ϕ*



Before moving forward, it is important to ask whether these new types of ϕ are built for purely phonological reasons: for instance, to avoid violations of MINIMAL BINARITY (which would emerge frequently if every postverbal argument were parsed into its own maximal ϕ). Fortunately, there is good evidence to suggest that they are not. The case runs as follows: there are cases where it would be possible to satisfy Minimal Binarity by building ϕ s that demonstrably lack correspondents in the syntax: for instance, ϕ s that contained both prosodically isolated arguments and adjuncts to the *voiceP*. When we set up the relevant configuration, however, we can see that these types of ϕ s are never built: in a VS clause where the verbal complex contains three ω s, a single- ω S will never be parsed into a $\phi_{[MAX]}$ with a following X (62b).

- (62) *Weight Manipulations: No Visible Eurythmic Effects*

- a. $\{\phi_{[MAX]}$ $\}$ $\{\phi_{[MAX]}$ $\}$
- | | | | |
|------|-------|----------------------|--------------------|
| máne | táma | mémaŋ ^H i | bálao ^H |
| mane | tama | memaj i | balao |
| just | go in | indeed | 3ABS |
| | | | mouse |

‘The mouse did indeed just run inside.’

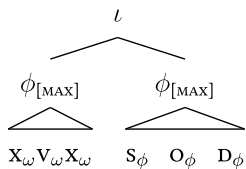
- b. $\{\phi_{[MAX]}$ $\}$ $\{\phi_{[MAX]}$ $\}$ $\{\phi_{[MAX]}$ $\}$
- máne táma mémaj^H i baláo^H díó^H
- mane tama memaj i balao dio
- just go in indeed 3ABS mouse there

‘The mouse did indeed just run inside there.’

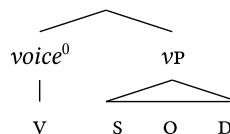
In light of this fact, I propose that ϕ s are built around the substrings SO, SD, OD, and SOD in direct response to the mapping constraint MAX-XP: namely, a category-blind pressure to translate every XP into a ϕ s. From this perspective, the Exposure Effect provides evidence for a headless functional projection that contains the S, O, and D. More specifically, I propose that it provides evidence for a headless ν P.

(63) *Prosodic Evidence for the ν P*

a. *The Phonology: SOD*



b. *The Syntax: SOD*



The following tableau shows the derivation of this effect, focusing on the antipassive VSO clause in (52b). In this clause, the mapping constraint MAX-XP demands the construction of ϕ s that correspond to the *voice*P and ν P (and plausibly the headless VP, not shown). Maximal Binarity then blocks the integrated parse in Candidate (a), where the four-word string forms a ϕ_{MAX} . The result is that the winner is the split parse in Candidate (b), where the PFX-V string forms a ϕ_{MAX} that plausibly corresponds to the *voice*P and the SO string forms a ϕ_{MAX} that corresponds to the ν P.

(64) *Deriving the Exposure Effect*

$[voiceP \text{ } \widehat{manj}\text{-}v^0 \text{ } [\nu P \text{ } [_{DP} \text{ } S] \text{ } [\nu P \text{ } [_{NP} \text{ } O]]]]]$	MATCH(ϕ ,XP)	MAX-BIN
a. $\{\phi_{[MAX]} \text{ } voiceP \text{ } \widehat{manj}\text{-}v \text{ } \{\phi \text{ } \nu P \text{ } S \text{ } O \text{ } \}\}$		*!
b. $\{\phi_{[MAX]} \text{ } voiceP \text{ } \widehat{manj}\text{-}v \text{ } \{\phi_{[MAX]} \text{ } \nu P \text{ } S \text{ } O \text{ } \}\}$		

We can now leverage the same manipulation to build a case for the *appl*P. By adding ω s in the right places in a VSOD string, we can create a prosodic configuration in which the substring OD cannot be parsed into a higher ϕ with the preceding V or S. The following example shows what happens when this is done: the O and the D are visibly parsed into a maximal ϕ .

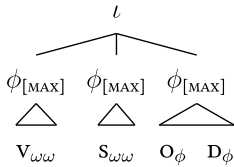
- (65) $\{\phi_{[MAX]}$ $\}$ $\{\phi_{[MAX]}$ $\}$ $\{\phi_{[MAX]}$ $\}$ $\{\phi_{[MAX]}$ $\}$
- máne nalattári^H i solána^H jáli^H do taláye^H xúru^H ó^H
- mane na-lattári i sola-na iali do talagae guru o
- just 3ERG-throw 3ABS friend-3GEN NAME that tomato teacher there

‘Ali’s friend just threw that tomato there at the teacher!’

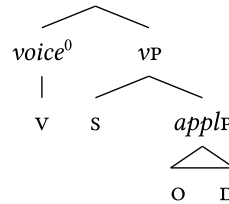
The emergence of this ϕ , in (66a), reveals the presence of the *applP* in (66b).

(66) *Prosodic Evidence for the applP*

a. *The Phonology: OD*



b. *The Syntax: OD*



Stepping back, then, these results reinforce the hypothesis that the syntax-prosody interface can build ϕ s in correspondence with all types of XPs at the moment where syntax and phonology meet (Selkirk 2009, 2011; Elfner 2012, 2015). In doing so, they adduce further evidence that functional XPs can be translated to ϕ s (Elfner 2015; Itô and Mester 2019a). On the analysis of clause structure adopted here, they also reveal that the mapping algorithm can build ϕ s around XPs whose heads have been carried away by word-building head-movement (cf. Kalivoda and Bellik 2021; Van Handel 2021). In this connection, they also reveal that such XPs need not be destroyed by head-movement of this type (cf. É Kiss 2008; Stepanov 2012).

Turning back to the syntax, these results provide an immediate rejoinder to our earlier observation on constituency: while it is difficult to detect headless VP-shells in the VSOD string in the syntax, it is possible to isolate and reveal these constituents in the prosody. This result opens up a novel line of evidence that the extended VP is built from a nested series of functional projections, and in doing so, it reconfirms the guiding hypothesis of a literature that has reached the same conclusion on many different grounds.

5 The syntax and prosody of scrambling

From the facts of surface prosody, we have reverse-engineered the fine syntactic constituency of the VSOD string. The final task that remains is thus to address the ordering alternations of Sect. 2—and more specifically, the syntax of deviant orders like VOS.

We can begin with a return to the linear position of VP-level adjuncts. We have seen that the arguments that surface outside the order S-O-D must follow all overt adjuncts to the VP (setting aside the clause-final *e* ‘here’ and *o* ‘there’). When an X is introduced in a VOS clause, for instance, it must precede the misordered S (67).

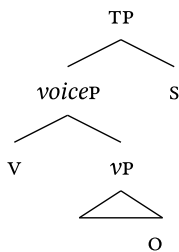
(67) *Misordered Arguments follow Final Adjuncts to the VP*

Na-itai i [_O iCicci’] [_X dio] [_S iKaco’].
 3ERG-seek 3ABS NAME there NAME

‘Kacho’ is looking for Chichi’ there.’

In light of this fact, I propose that these ordering alternations involve a process of rightward scrambling. This process shifts arguments to a position outside the *voiceP* that is linearized to the right of all Xs, yielding the syntax in (68) for a VOS string.

(68) *VOS* → *Rightward Scrambling of S*



This analysis breaks from many alternatives that would derive orders like VOS from movement to the left. Two such alternatives stand out: one that derives VOS from leftward movements of the V and O (Otsuka 2005); (69a) and one that derives VOS from the fronting of a constituent like the VP (Massam 2001a,b); (69b).

(69) *Alternative Analyses of VOS*

- a. LEFTWARD SCRAMBLING: $[\text{voiceP } V [\text{FP } O [\text{VP } S [\text{VP } __]]]]$
- b. PREDICATE-FRONTING: $[\text{voiceP } [\text{VP } V O] [\text{VP } S [\text{VP } V __]]]$

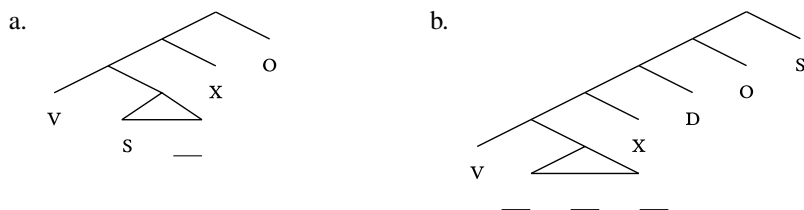
Adopting the rightward scrambling analysis for Mandar, many facts fall into place. First, this analysis explains the linear position of VP-level adjuncts: if these elements are right-adjoined to the *voiceP*, then they should follow the S, O, and D when they remain in the *voiceP* (70a) and precede them when they are scrambled out (70b).

(70) *The Position of Adjuncts*

- a. WITHOUT SCRAMBLING: $[\text{voiceP } [\text{voiceP } V [\text{VP } S O]] X]$
- b. WITH SCRAMBLING: $[\text{FP } [\text{voiceP } [\text{voiceP } V [\text{VP } __ O]] X] S]$

Second, this analysis allows us to explain every attested order of arguments in the postverbal space. The order VOS arises from rightward scrambling of the S. Other strings with one misordered argument, like VSXO, are derived by rightward scrambling of the misordered argument (71a). Strings that contain multiple misordered arguments, like VXDOS, are derived by scrambling multiple arguments (71b).

(71) *Deviations from VSOD*

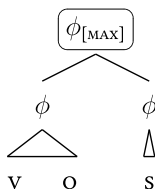


5.1 The prosody of scrambling

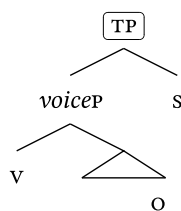
These syntactic results set up a final prediction on the prosody of strings with orders like VOS. If the misordered arguments in these strings are scrambled out of the *voiceP*, then we should expect to see the parse in (80b): a ϕ should be built around the *voiceP* and then a higher ϕ should be built around the *voiceP* and the arguments that have been scrambled out, corresponding to a higher functional projection like the TP (80a).

(74) *The Expected Prosody of Scrambling*

a. *The Expected Prosody*



b. *The Expected Syntax*



This prediction is incorrect: when a V-NP-NP string forms a single maximal ϕ under default prosodic conditions, it cannot receive an interpretation like VOS. Instead, the only possible interpretations are those that conform to the order VSOD. The following example illustrates with a transitive V-NP-NP string: under the parse $\{\phi_{[MAX]} \text{ V NP NP} \}$, this clause must receive the interpretation VSO.

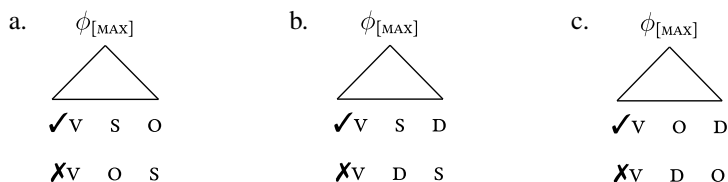
(75) *Phrasing and Interpretation*

$\{\phi_{[MAX]} \}$				$\} \{\phi_{[MAX]} \}$	
néte ^H	i	irípe ^H	io	áli ^H	ó ^H
na-itai	i	iripa?i	do	ali	o
3ERG-look for 3ABS NAME that NAME				there	

ONLY VSO: ‘Ripa’i is looking for that Ali there.’

The following diagram summarizes this constraint: when a three-word V-NP-NP string is parsed into a maximal ϕ , it must be interpreted as VSO, VSD, or VOD (76).

(76) *Prosodic Constituency to Interpretation*



VOS, and (ii) they require transitive V-NP-NP-X strings to receive the interpretation VSOX. It thus seems natural to posit that, in the clauses that contain transitive verbs in these languages, VSO is the unmarked order, VSO strings can form intact *voicePs*, and VOS order is derived through rightward scrambling of the S.

Scrambled arguments must form independent maximal ϕ s. The following example illustrates: in a three- ω VOS string, the V and O form a ϕ_{MAX} that excludes the S.

(77) *Misordered S $\rightarrow$ Maximal ϕ*

{ ϕ_{MAX} }			{ ϕ_{MAX} }		{ ϕ_{MAX} }
néte ^H	i	iripáʔi ^H	do	áli ^H	ó ^H
na-itai	i	iripaʔi	do	ali	o
3ERG-look for	3ABS	NAME	that	NAME	there

‘That Ali there is looking for Ripa’i.’

The same parse extends to all types of scrambled arguments. Example (78a) shows this effect in a VDS clause, where the S is scrambled and is once again forced to form its own ϕ_{MAX} . Example (78b) shows the same fact in a VDO clause, where the final O has been scrambled and is thus forced to form a ϕ_{MAX} as well.

(78) *Misordered Arguments $\rightarrow$ Maximal ϕ s*

a. { ϕ_{MAX} }

najolóa ^H	ã	iripáʔi ^H	do	áli ^H	ó ^H
na-dʒolo-aŋ	aʔ	iripaʔi	do	ali	o
3ERG-point-APPL	1ABS	NAME	that	NAME	there

‘That Ali there showed me to Ripa’i.’

b. { ϕ_{MAX} }

ujolóa ^H	ĩ	iripáʔi ^H	do	áli ^H	ó ^H
u-dʒolo-aŋ	i	iripaʔi	do	ali	o
1ERG-point-APPL	3ABS	NAME	that	NAME	there

‘I showed that Ali there to Ripa’i.’

The facts are the same when multiple arguments are scrambled to the right. In a VDOS clause where both the O and the S are scrambled to the right, for instance, each of those argument is forced to form its own maximal ϕ .

(79) *Multiple Misordered Arguments $\rightarrow$ Multiple Maximal ϕ s*

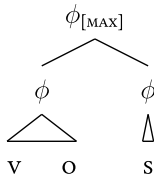
{ ϕ_{MAX} }		{ ϕ_{MAX} }		{ ϕ_{MAX} }	
nalattʃári ^H	i	iripáʔi ^H	talayáe	do	áli ^H
na-lattʃari	i	iripaʔi	talagae	do	ali
3ERG-throw	3ABS	NAME	tomato	that	NAME
				there	there

‘That Ali there threw the tomato at Ripa’i.’

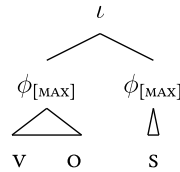
We thus arrive at a final prosodic puzzle: the mapping constraints of Match Theory predict that scrambled arguments should form ϕ s with the *voiceP* (80a), but, in fact, scrambled arguments always form maximal ϕ s of their own (80b).

(80) *The Prosodic Result of Scrambling*

a. *The Expected Parse*



b. *The Attested Parse*

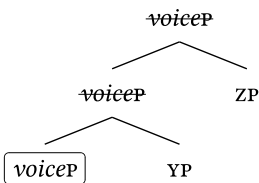


5.2 Scrambling and adjunction

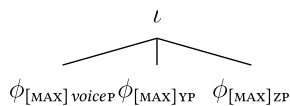
I propose that the solution to this puzzle lies in a connection to adjunction. In Sect. 2, we saw that VP-level adjuncts are always each forced to form their own maximal ϕ s. We can begin to understand this pattern by drawing a connection to a proposal from Truckenbrodt (1999): in adjunction structures of the shape $[_{XP} [_{XP} XP] YP]$, where YP is an adjunct adjoined to XP, the mapping algorithm maps the lower segment of XP to a ϕ and ignores higher segment of the XP that is created by adjunction (May 1985). This proposal yields the mapping in (81): if two Xs are adjoined to the *voiceP*, then a ϕ will only be built around the lowest segment of the *voiceP* (which is boxed below) and the following VP-level adjuncts will each form independent ϕ s.

(81) *The Mapping of Adjunction Structures*

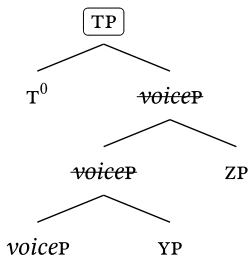
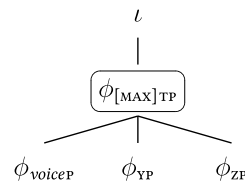
a. *VoiceP: Input*



b. *VoiceP: Output*



It is natural that adjunction should be singled out in the prosody in such a way, as the argument-adjunct distinction is fundamental to the syntax and is also preserved at the interface with LF (Lebeaux 1988). As such, Selkirk (2011, 483 fn.38) introduces the proposal of Truckenbrodt (1999) into Match Theory. To derive the result that we want, however, we will have to go slightly farther than this. This is because Match Theory holds that all XPs should form ϕ s when phonological conditions allow, and as a result, it predicts that the *voiceP* and all following Xs may still be parsed into a larger ϕ if they are dominated by a higher functional projection that only contained three ω s: for instance, the boxed TP in tree (82a).

(82) *A Problem for Match Theory*a. *TP: Input*b. *Unattested Output*

To derive the parse of adjuncts, then, we must find a way to force them to form maximal ϕ s. Most versions of Match Theory do not contain constraints of this type, which force the mapping of specific XPs to relational subtypes of ϕ s like the $\phi_{[\text{MAX}]}$ (though cf. Ishihara 2014, and see Van Handel et al. 2024 for discussion). As a result, I propose that this restriction must emerge from the constraint in (83): one which demands that XP-level adjuncts be excluded from every higher ϕ that contains the constituent to which they adjoin.^{14,15}

(83) *On the Prosodic Mapping of Phrasal Adjuncts*

REPEL: Let SX be an input syntactic representation and PR its corresponding output representation. Assign one violation for every adjunction structure of the shape $[_{XP} [_{XP} XP] YP]$ in SX, YP a phrasal adjunct, for which the output correspondents of XP and YP are contained within a single ϕ in PR.

Ranked above MATCH(ϕ ,XP), this constraint forces VP-level adjuncts to be parsed into separate maximal ϕ s from the *voiceP*, as shown in the following tableau.

(84) *Deriving the Prosodic Parse of Adjuncts*

$[_{TP} [_{voiceP} [_{voiceP} V] [_{YP} \text{HERE}]]]$	REPEL	MATCH(ϕ ,XP)
a. $\{\phi_{[\text{MAX}]TP} V \text{ HERE} \}$	*!	
b. $\{\phi_{[\text{MAX}]TP} V \} \{ \phi_{[\text{MAX}]YP} \text{HERE} \}$		*

¹⁴ X^0 -level adjunction is subject to a similar constraint. There is a prosodic difference between the complex X^0 s that are built through Amalgamation (word-building head-movement; Harizanov and Gribanova 2019) and those which are built by adjoining X^0 s to X^0 s (for instance, N^0 - ADJ^0 sequences): in general, the former are parsed into single ω s and the latter are parsed into two. This suggests that adjunct X^0 s are parsed in line with an X^0 -level version of REPEL, which demands that structures with the shape $[_X [_X X] Y]$, Y a head-level adjunct, not be parsed into one ω . Naturally, the resultant structure must be parsed into a ϕ (just as the result of XP-adjunction must be parsed into an ι); see Bellik and Kalivoda (2016) on this point.

¹⁵ REPEL must be a violable constraint, as its effects can be overridden by the phonology of focus (ϕ s can be built around strings like VSX in Mandarin when the X is contrastively focused). There may also be XP-level adjuncts that evade this constraint, like the VP-final adverbs that carry nuclear stress in English (e.g. *well*; see Cinque 1993 for discussion). Much further work is required on both the syntax and the interface constraints that lie beneath these effects.

In clauses that contain multiple VP-level adjuncts, each adjunct can be forced into an independent maximal ϕ by the ranking of Minimal Binarity beneath DEP- ϕ , which bans the construction of ϕ s that do not correspond to XPs (Itô and Mester 2019a).

(85) *Deriving the Parse of Multiple Adjuncts*

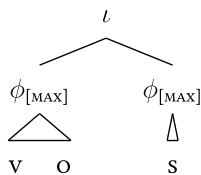
- a. DEP- ϕ : Let SX be an input syntactic representation and PR its corresponding output representation. Assign one violation for every ϕ in PR which does not correspond to an XP in SX.

	$[_{TP} [_{voiceP} [_{voiceP} [_{voiceP} V] [_{VP} \text{ HERE }]] [_{ZP} \text{ TODAY }]]]$	DEP- ϕ	MIN-BIN
a.	$\{\phi_{[MAX]_{TP}} V \} \{\phi_{[MAX]} \text{ HERE TODAY } \}$	*!	*
b.	$\{\phi_{[MAX]_{TP}} V \} \{\phi_{[MAX]_{VP}} \text{ HERE } \} \{\phi_{[MAX]_{ZP}} \text{ TODAY } \}$		***

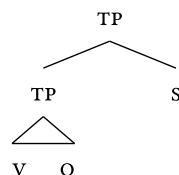
This analysis opens up a path to derive the attested parse of scrambled arguments: we can force these constituents to form maximal ϕ s by subjecting them to REPEL. On this analysis, the aberrant prosody of scrambled arguments follows from a fact about their syntax: scrambled arguments occupy adjunct positions in the surface syntax. The following trees illustrate: in a clause that contains a scrambled DPS, like a clause of the order VOS, the scrambled DP forms a maximal ϕ because it is adjoined to the TP.

(86) *Scrambling $\rightarrow$ Adjunction*

a. TP: Prosody



b. TP: Syntax



The following tableau shows the derivation of this effect: in a VOS clause, the S is forced into its own maximal ϕ by the ranking of REPEL over MATCH(ϕ ,XP).

(87) *Deriving the Parse of Scrambled Arguments*

	$[_{TP} [_{TP} V O] [_{DP} S]]$	REPEL	MATCH(ϕ ,XP)
a.	$\{\phi_{[MAX]_{TP}} V O \} \{\phi_{[MAX]_{DP}} S \}$		*
b.	$\{\phi_{[MAX]_{TP}} \{\phi_{TP} V O \} \{\phi_{DP} S \} \}$	*!	

This result takes on syntactic importance in tandem with a second fact: this type of scrambling shows the binding profile of $\bar{A}$ -movement. The following examples illustrate. Example (88a) presents a baseline fact about Condition C of the Binding Theory (Chomsky 1981; Reinhart 1983): in Mandarin, the ditransitive O cannot contain an R-expression that is coindexed with a pronominal S. Example (88b) then shows that scrambling has no effect on this constraint: a ditransitive O still cannot contain an R-expression that is coindexed with a pronominal S when it is scrambled to the right and placed outside the *voiceP*.

(88) *Scrambling: Reconstruction for Condition C*

- a. Na-na-bengang o [s *pro*] [o buku na-alli iNina] marondong.
 will-3ERG-give 2ABS *pro* book 3ERG-buy NAME tomorrow
 ‘Tomorrow she_{*i,j} will give you the book that Nina_i bought.’
- b. Na-na-bengang o [s *pro*] t_o marondong [o buku na-alli iNina].
 will-3ERG-give 2ABS *pro* tomorrow book 3ERG-buy NAME
 ‘Tomorrow she_{*i,j} will give you the book that Nina_i bought.’

In this respect, scrambling contrasts with movement of the absolutive argument to its high clause-internal A-position. This second process does affect Condition C: in a transitive clause where the O is absolutive and raises to this A-position covertly, the O can contain an R-expression that is coindexed with a pronominal S (89).

(89) *Absolutive Movement to Spec,TP: No Reconstruction for Condition C*

- Na-na-baca i [s *pro*] [o buku na-alli iNina] marondong.
 will-3ERG-read 3ABS *pro* book 3ERG-buy NAME tomorrow
 ‘Tomorrow she_{i,j} will read the book that Nina_i bought.’

The following diagram illustrates this split. As scrambling does not influence the calculus of Condition C, it must reconstruct (90a). As movement to SPEC,TP feeds the calculus of Condition C (and influences patterns of variable binding), in contrast, it must avoid reconstruction (90b). The copies that are interpreted are boxed below; the copies that are not interpreted are struck through.

(90) *Asymmetries in Reconstruction*

- a. SCRAMBLING: [TP [TP [voiceP V [vP S [vP O]]]]] $\bar{\Theta}$.
 $\downarrow$
- b. ABSOLUTIVE MOVEMENT: [TP O [voiceP V [vP S [vP $\bar{\Theta}$]]]]].
 $\downarrow$

These observations suggest that this process of scrambling must be an $\bar{A}$ -movement. From this position, we can leverage the prosody of scrambling to make a further claim: in Mandarin, $\bar{A}$ -scrambling must place its targets in adjunct positions. The result is a prosodic case for a classical syntactic hypothesis: $\bar{A}$ -scrambling places its targets in adjunct positions (Chomsky 1993; Müller and Sternefeld 1994, 1996).

We can now address the specific syntax of this operation. If scrambling is driven by EPP features on an attracting x^0 (e.g., Ko 2005, 2007), then it should not place its targets in adjunct positions. This is because EPP features are essentially selectional and should thus place their targets in selected specifiers (Chomsky 2001). In Mandarin, then, this type of scrambling cannot be driven by an attracting x^0 . I thus propose that it is driven by the needs of the scrambling elements themselves: in other words, it is driven by the principle of Greed (Lasnik 1995; Bošković 1995).

I propose that this operation of rightward $\bar{A}$ -scrambling is driven by a feature on a null definite D^0 : a selectional feature that forces the DP to move to a position where this feature can be checked through adjunction to the TP (on the hypothesis that adjuncts select for their hosts, Bruening 2013). This feature is presented in (91a), using

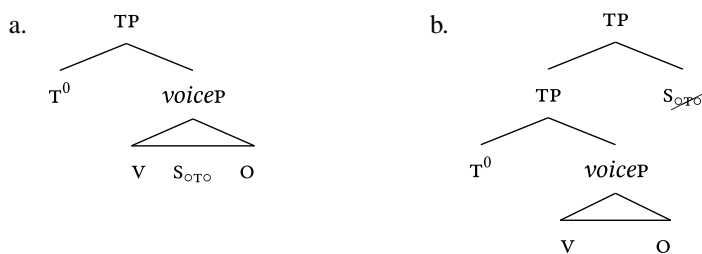
the ring notation ($\circ$) of Zyman (2025). When this D^0 combines with a complement, the resultant DP will be forced to move out of the *voiceP*. The same movement is not forced by the other null definite D^0 in the language, which lacks this feature (91b).

(91) *Two Definite Determiners in Mandarin*

- a. D^0_{+DEF1} : [$\circ$ To $\circ$]
- b. D^0_{+DEF2} : []

I assume that all definite arguments can combine with either of the two null definite D^0 s in (91) in a free lexical choice. When the S merges with the regular D^0 , it will stay in the *voiceP*, yielding an eventual linearization of VSO. When it merges with the scrambling D^0 , as in (92a), it will scramble to satisfy the selectional feature on that D^0 and eventually be linearized to the right of its host, yielding the order VOS (92b).

(92) *The Alternation Again*



The result is an analysis that captures the defining properties of this process in Mandarin: it is always optional, it is always able to target any number of arguments in a clause, and it can place them in any order after the *voiceP*. In tandem with this syntactic success, this analysis fits naturally with the prosodic signature of this operation, explains how it might place its targets in adjunct positions, and falls in line with the larger hypothesis, advanced by Chomsky et al. (2019), that scrambling is “not head-oriented in any plausible sense.”

6 Conclusion

This investigation of syntax and prosody has led us to the understanding that VSOD strings in Mandarin typically form syntactic constituents that are built from a series of functional projections, with their linear order derived from word-building head-movement of the V and lower-copy spell-out of absolutive arguments that raise to SPEC,TP. Our syntactic results have also led us to posit a process of rightward A-scrambling to derive deviant orders like VOS, and the prosody has suggested that this process places its targets in adjunct positions. To the extent that this analysis is successful, it reinforces four architectural points. First, the extended VP must contain an internal sequence of functional projections, even when head-movement carries the V along its spine. Second, A-chains must be reduced in a calculus that does not mandate higher-copy spell-out. Third, adjuncts and specifiers must be able to be linearized to

the right of their hosts. Fourth and finally, one case of $\bar{A}$ -scrambling must place its targets in adjunct positions and may thus be driven by Greed.

At the interface of syntax and prosody, our results add to the emerging consensus that prosodic phrasing can be leveraged to preserve functional constituency. Our results also reinforce the hypothesis that adjunction structures are phonologized in a specific and regular way, yielding a prosodic distinction between the structures built by MERGE and ADJOIN. The narrow force of this result is to deliver a surface-oriented diagnostic for the argument-adjunct distinction; the larger implication is that the interface may leverage prosodic phrasing to preserve other syntactic relationships in similarly distinctive ways.

In service of this final point, we can conclude our investigation with a final note on the prosody of selection. In Sect. 4, we leveraged a weight constraint to prosodically split the V from its arguments and study the prosodic organization of the postverbal space. But there is a context in which the introduction of additional words forces a parse that reflects neither the syntax we have discovered nor the phonological pressures we have identified so far: when a one- ω V is followed by three one- ω arguments, the VSOD string must be parsed $\{\phi_{[\text{MAX}]}\text{ VS}\} \{\phi_{[\text{MAX}]}\text{ OD}\}$.

(93) *The Aberrant Parse of VSOD*

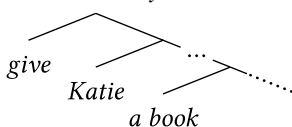
$\{\phi_{[\text{MAX}]}\}$			$\{\phi_{[\text{MAX}]}\}$			$\{\phi_{[\text{MAX}]}\}$
nawéŋa ^H	ĩ	iripáʔi ^H	do	wúku ^H	yúru ^H	ó ^H
na-be-ŋaŋ	i	iripaʔi	do	buku	guru	o
3ERG-give-APPL 3ABS NAME			that	book	teacher	there

‘Ripa’i gave the teacher that book there.’

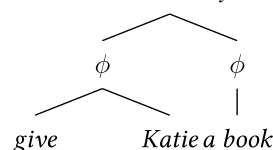
The emergence of this parse reflects a larger tendency at the syntax-prosody interface whose systematicity was first discovered by Kalivoda (2018): when weight constraints block the construction of a ϕ around a selecting X^0 and its arguments, but do not force the selecting X^0 to form a maximal ϕ of its own, then the selecting X^0 is often parsed into a ϕ with the closest argument alone. I will refer to this pattern as Kalivoda’s Effect. Kalivoda (2018) observes that it can be seen at least in three- ω vso strings in Irish (Elfner 2015) and in V-DP-DP ditransitives in a much wider range of languages. Kalivoda (2018) also notes that it yields a syntax-prosody mismatch in the English VP: the DP-DP ditransitive shows the descending syntax in (94a) (Barss and Lasnik 1986) but the ascending prosody in (94b) (Hayes 1989; Elfner 2014).

(94) *Kalivoda’s Effect: The English Ditransitive*

a. *Ditransitive Syntax*



b. *Ditransitive Prosody*



Kalivoda (2018) proposes that the pressure that underlies this effect is bound up with the phonologization of C-command. But as there is a regular phonological reflex to adjunction, it is natural to wonder whether Kalivoda's Effect may reveal a parallel phonological reflex to selection: more specifically, a phonological mirror to REPEL which demands that the *v* be parsed into a ϕ with every element that it selects (cf. Clemens 2016, 2019). The patterns of mismatch that emerge through Kalivoda's Effect, like $\{\phi_{[MAX]} \text{ VS } \} \{\phi_{[MAX]} \text{ OD } \}$, would then be driven by the interaction of weight constraints with this specific constraint.

(95) *On the Mapping of Selected Arguments*

ATTRACT: Let SX be an input syntactic representation and PR its corresponding output representation. Assign one violation for every structure of the shape $[_{XP} X^0 [_{YP} YP]]$ in SX , X^0 selecting YP , for which the output correspondents of X^0 and YP are not contained in a ϕ in PR .

Whatever the analysis of Kalivoda's effect, its existence suggests that the prosody of the extended VP must emerge from the interaction of three types of pressure: requirements for syntax-prosody isomorphism, output-oriented restrictions on prosodic well-formedness, and interface constraints that force particular syntactic relationships to be prosodified in regular ways. It will be essential to study each of these components as we advance on the prosody—and thus the syntax—of the extended VP.

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